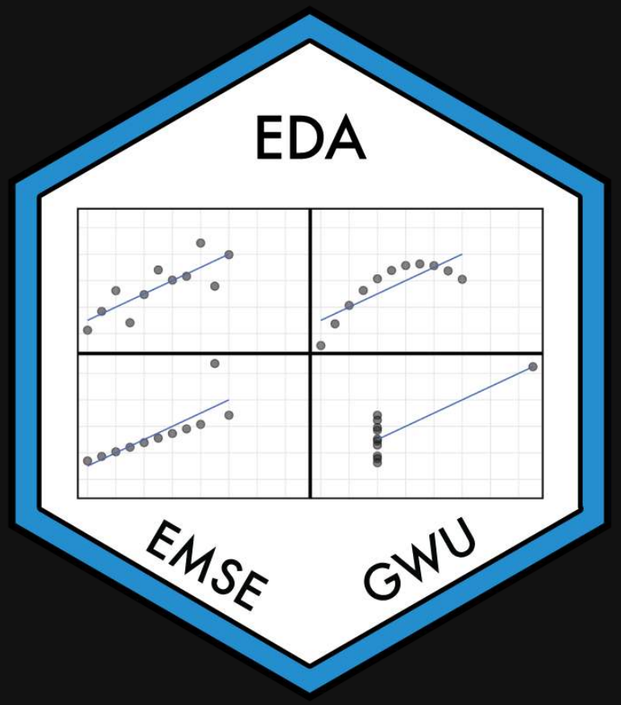


Maps

EMSE 4572/6572: Exploratory Data Analysis

John Paul Helveston

November 11, 2026

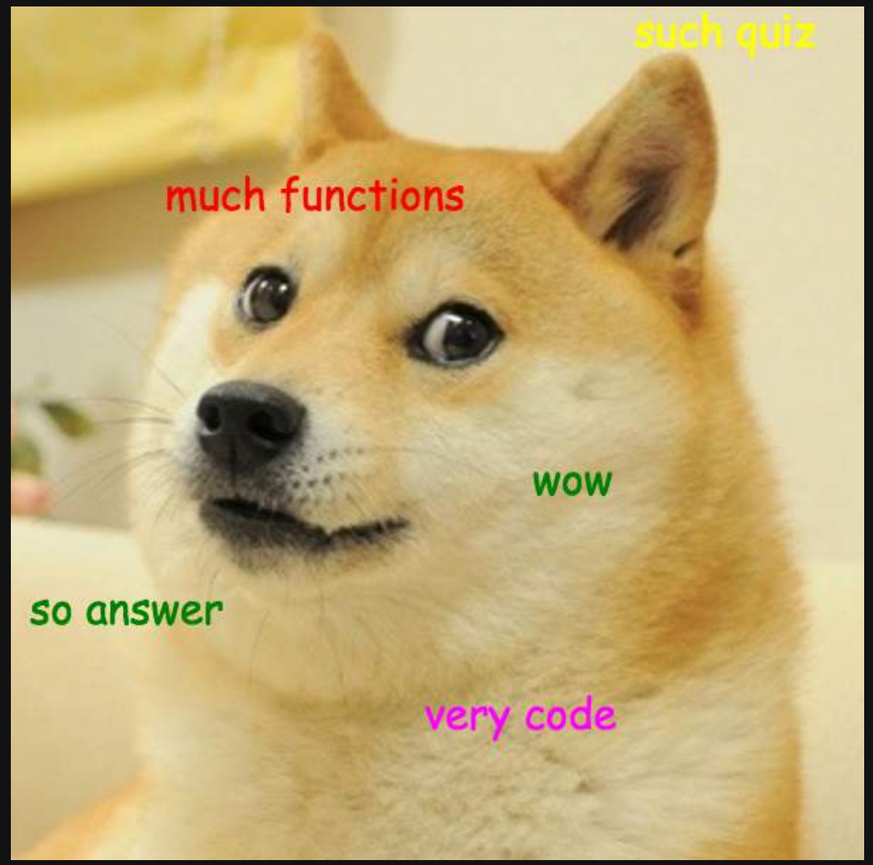
Quiz 5 - last one!

12:00

Write your name on the quiz!

Rules:

- Work alone; no outside help of any kind is allowed.
- No calculators, no notes, no books, no computers, no phones.



Today's data

```
milk_production <- read_csv(file.path('data', 'milk_production.csv'))
us_coffee_shops <- read_csv(file.path('data', 'us_coffee_shops.csv'))
```

New packages:

```
install.packages('maps')
install.packages('mapproj')
install.packages('sf')
install.packages('rnaturalearth')
devtools::install_github("ropensci/rnaturalearthhires")
devtools::install_github("ropensci/rnaturalearthdata")
```

Week 12: *Maps*

1. Plotting maps
2. Adding data to maps
3. Projections

Week 12: *Maps*

1. Plotting maps
2. Adding data to maps
3. Projections

How to make a map

Step 1: Load a shape file

a. Use a library
b. Read in a shape file

Step 2: Plot the shape file

a. Polygon data: `geom_polygon()`
b. Simple Features data: `geom_sf()`

Polygon maps

Get the “World” shape file

```
1 library(ggplot2)
2
3 world <- map_data("world")
4 head(world)
```

```
#>         long      lat group order region subregion
#> 1 -69.89912 12.45200     1     1  Aruba      <NA>
#> 2 -69.89571 12.42300     1     2  Aruba      <NA>
#> 3 -69.94219 12.43853     1     3  Aruba      <NA>
#> 4 -70.00415 12.50049     1     4  Aruba      <NA>
#> 5 -70.06612 12.54697     1     5  Aruba      <NA>
#> 6 -70.05088 12.59707     1     6  Aruba      <NA>
```

Polygon maps

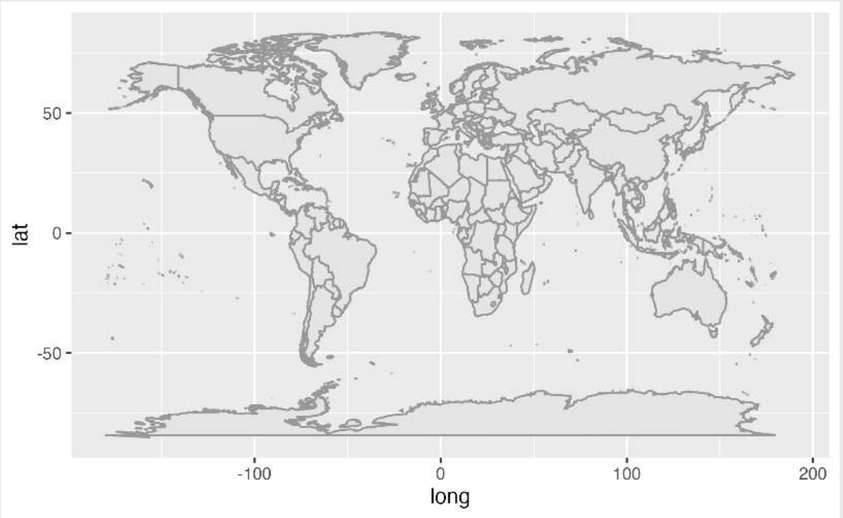
Get the “World” shape file

```
library(ggplot2)

world <- map_data("world")
```

Make the plot with `geom_polygon()`

```
ggplot(world) +
  geom_polygon(
    aes(x = long, y = lat, group = group),
    fill = "grey90", color = "grey60"
  )
```



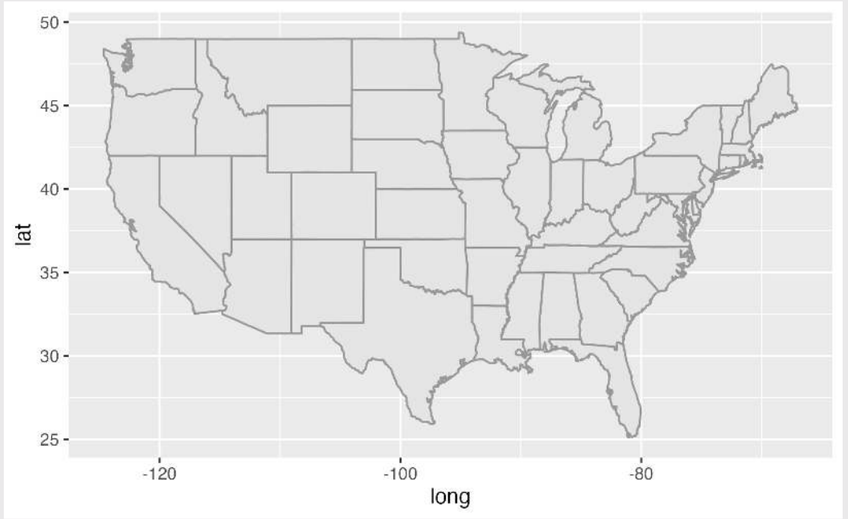
Polygon maps

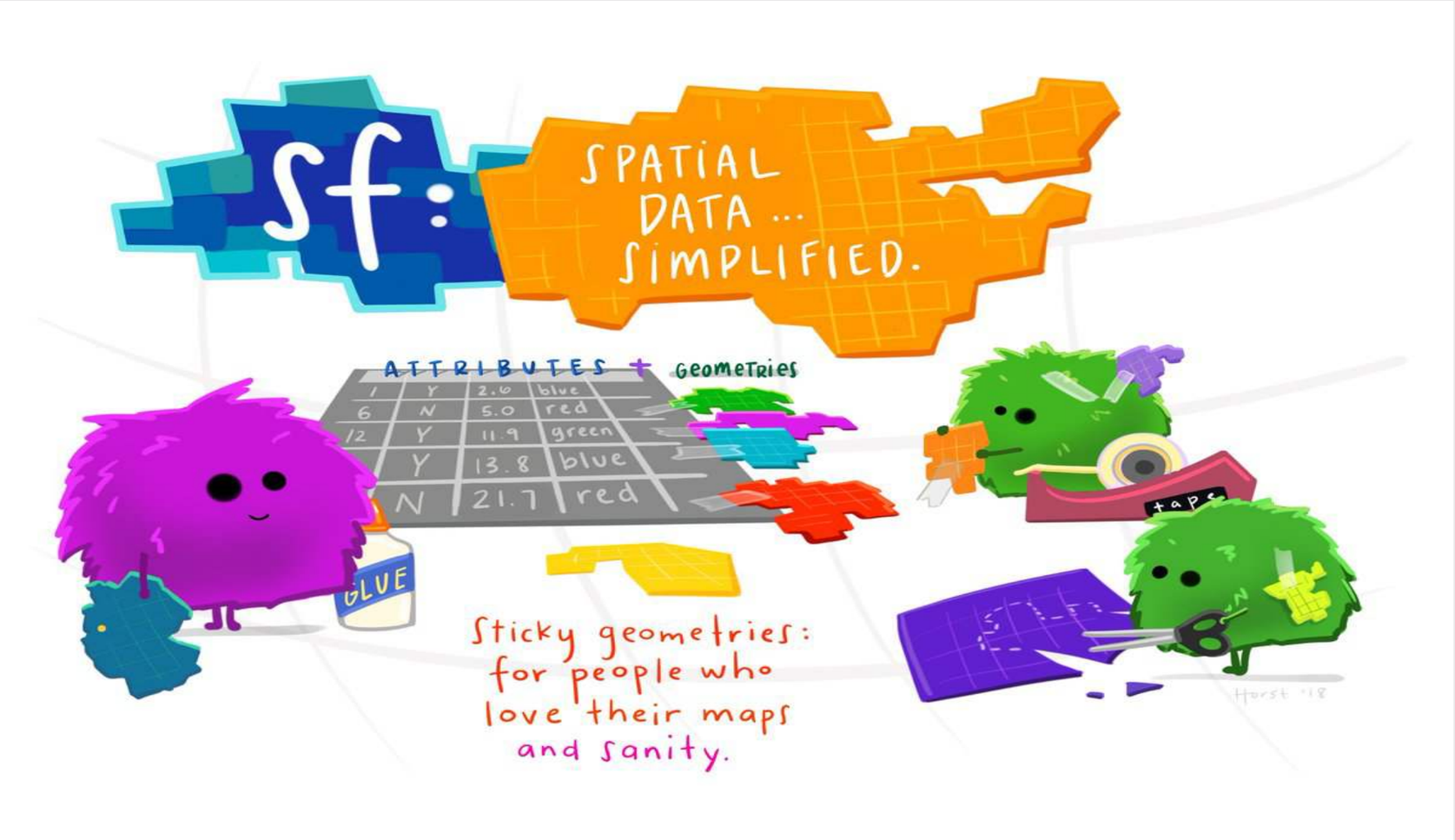
Get the “US States” shape file

```
1  library(ggplot2)
2
3  us_states <- map_data("state")
```

Make the plot with `geom_polygon()`

```
1  ggplot(us_states) +
2    geom_polygon(
3      aes(x = long, y = lat, group = group),
4      fill = "grey90", color = "grey60"
5    )
```





Simple Features package: {sf}

Art by Allison Horst

Simple Features (sf) maps

Library data from Natural Earth

```r
library(rnaturalearth)
library(rnaturalearthdata)

world <- ne_countries(
  scale = "medium",
  returnclass = "sf"
)

world %>%
    select(name, geometry) %>%
    head()
```

```
#> Simple feature collection with 6 features and 1 field
#> Geometry type: MULTIPOLYGON
#> Dimension:     XY
#> Bounding box:  xmin: -73.36621 ymin: -22.40205 xmax: 109.4449 ym
#> Geodetic CRS:  WGS 84
#>        name                          geometry
#> 1  Zimbabwe MULTIPOLYGON (((31.28789 -2...
#> 2    Zambia MULTIPOLYGON (((30.39609 -1...
#> 3     Yemen MULTIPOLYGON (((53.08564 16...
#> 4   Vietnam MULTIPOLYGON (((104.064 10....
#> 5 Venezuela MULTIPOLYGON (((-60.82119 9...
#> 6   Vatican MULTIPOLYGON (((12.43916 41...
```

Simple Features (sf) maps

Get the “World” shape file

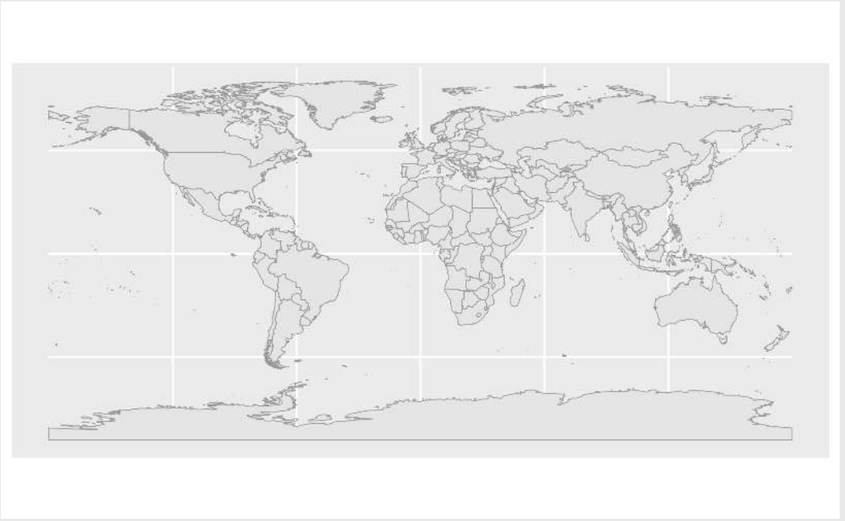
```r
library(rnaturalearth)
library(rnaturalearthdata)

world <- ne_countries(
  scale = "medium",
  returnclass = "sf"
)
```

Make the plot with `geom_sf()`

```r
library(sf)

ggplot(data = world) +
  geom_sf(fill = "grey90", color = "grey60")
```



Simple Features (sf) maps

Get the “US States” shape file

```
1 library(rnaturalearth)
2 library(rnaturalearthdata)
3
4 us_states <- ne_states(
5   country = 'united states of america',
6   returnclass = 'sf'
7 )
```

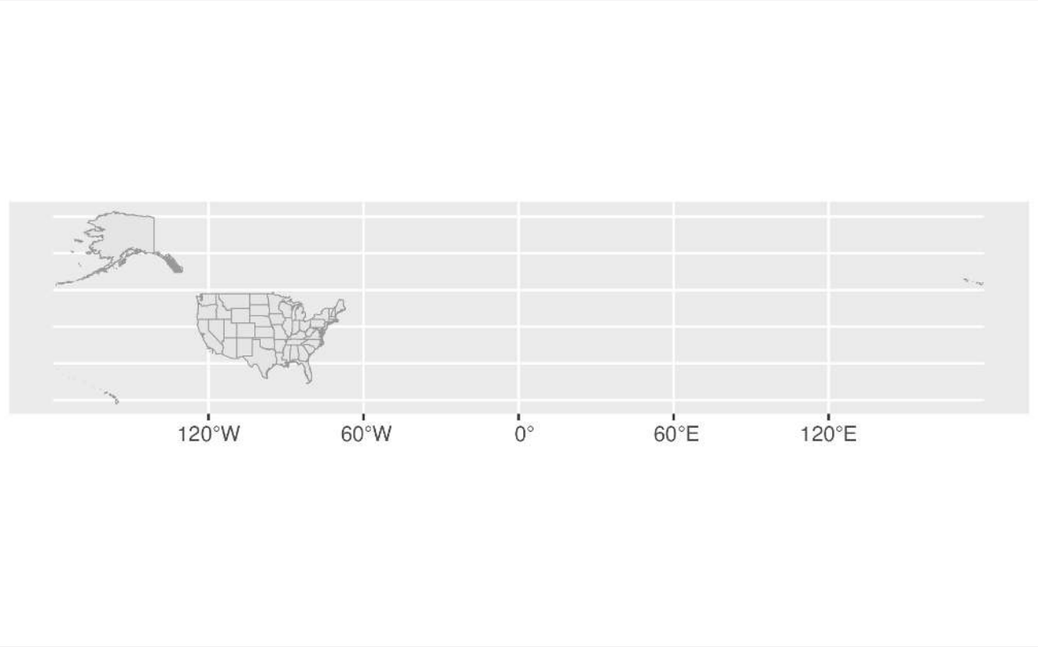
Simple Features (sf) maps

Get the “US States” shape file

```
1  library(rnaturalearth)
2  library(rnaturalearthdata)
3
4  us_states <- ne_states(
5    country = 'united states of america',
6    returnclass = 'sf'
7  )
```

Make the plot with `geom_sf()`

```
1  library(sf)
2
3  ggplot(data = us_states) +
4      geom_sf(fill = "grey90", color = "grey60")
```



Simple Features (sf) maps

Get the **Continental** “US States” shape file

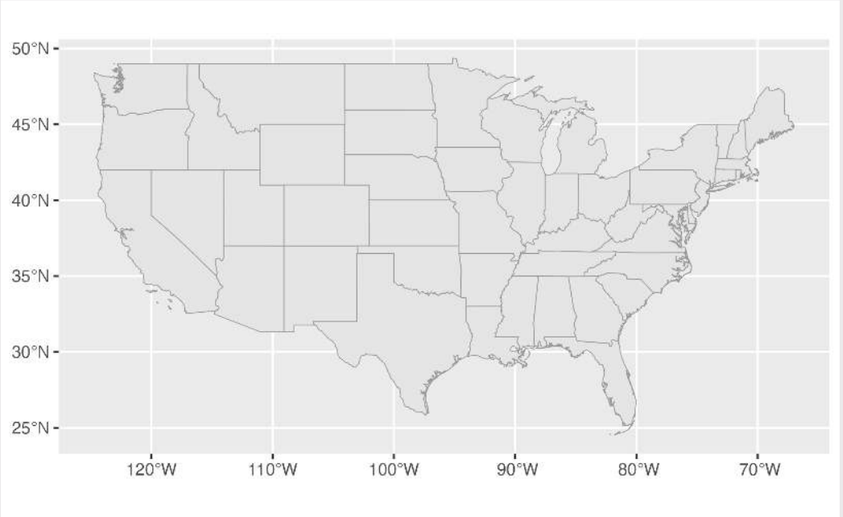
```
library(rnaturalearth)
library(rnaturalearthdata)

us_states_cont <- ne_states(
      country = 'united states of america',
      returnclass = 'sf'
    ) %>%
    filter(! name %in% c('Alaska', 'Hawaii'))
```

Make the plot with `geom_sf()`

```
library(sf)

ggplot(data = us_states_cont) +
  geom_sf(fill = "grey90", color = "grey60")
```



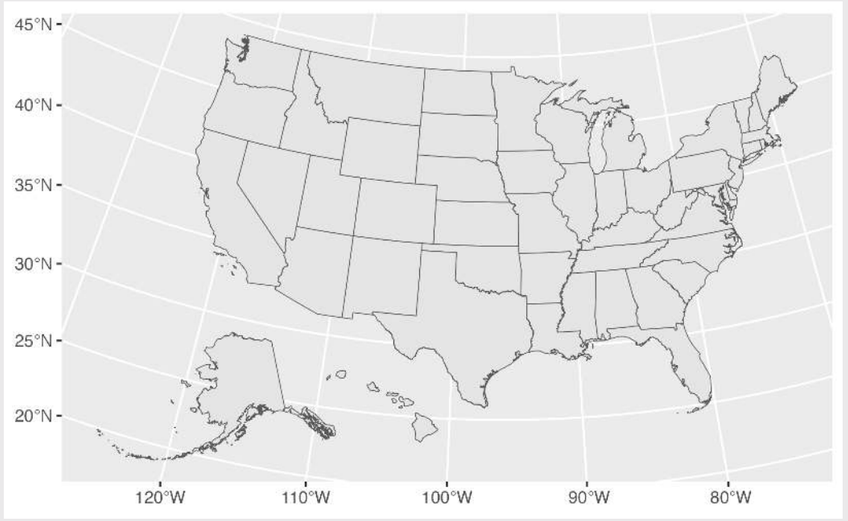
Inset Hawaii and Alaska

Get the shape file from {tigris} package

```
1  library(tigris)
2
3  us_sf <- tigris::states(class = "sf", cb = TRUE) %>%
4      shift_geometry() %>%
5      filter(GEOID < 60)
```

Make the plot with `geom_sf()`

```
1  us_sf %>%
2      ggplot() +
3      geom_sf()
```



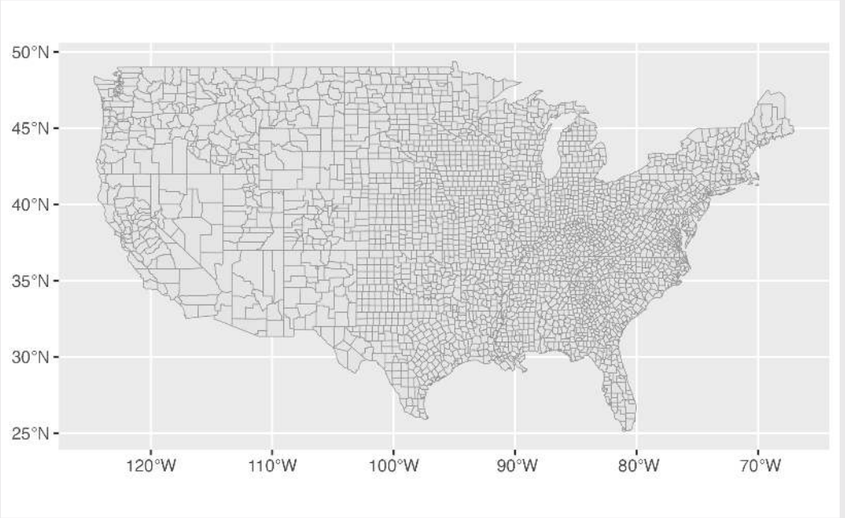
The `maps` package

Includes data on:

- World: world, world.cities, lakes
- US: states, county, state, usa
- France: france
- Italy: italy
- New zealand: nz

Example:

```
1  library(maps)
2
3  us_counties <- st_as_sf(
4      map("county", plot = FALSE, fill = TRUE))
5
6  ggplot(data = us_counties) +
7      geom_sf(fill = 'grey90', color = 'grey60')
```



Simple Features (sf) maps: `st_read()`

Read in the “World” shape file from Natural Earth

```
1 library(sf)
2
3 world <- st_read(file.path(
4     'data', 'natural_earth_countries',
5     'ne_50m_admin_0_countries.shp')) %>%
6     clean_names()
```

```
#> Reading layer `ne_50m_admin_0_countries' from data source `/User
#> Simple feature collection with 241 features and 94 fields
#> Geometry type: MULTIPOLYGON
#> Dimension:     XY
#> Bounding box:  xmin: -180 ymin: -89.99893 xmax: 180 ymax: 83.599
#> Geodetic CRS:  WGS 84
```

Simple Features (sf) maps: `st_read()`

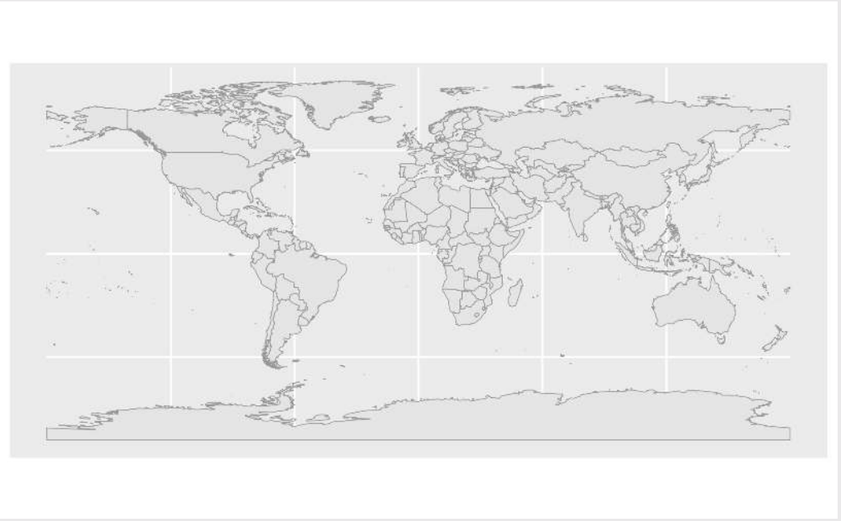
Read in the “World” shape file

```
library(sf)

world <- st_read(file.path(
  'data', 'natural_earth_countries',
  'ne_50m_admin_0_countries.shp')) %>%
  clean_names()
```

```
#> Reading layer `ne_50m_admin_0_countries' from data source `/User
#> Simple feature collection with 241 features and 94 fields
#> Geometry type: MULTIPOLYGON
#> Dimension:     XY
#> Bounding box:  xmin: -180 ymin: -89.99893 xmax: 180 ymax: 83.599
#> Geodetic CRS:  WGS 84
```

```
ggplot(data = world) +
  geom_sf(fill = "grey90", color = "grey60")
```



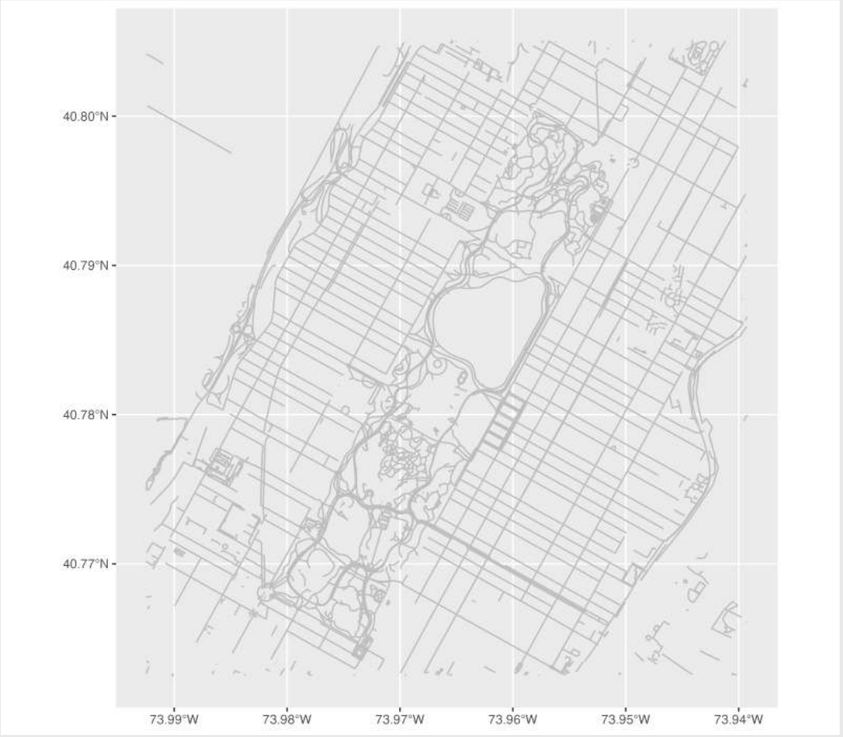
Simple Features (sf) maps: `st_read()`

Read in the “Central Park” shape file [source]

```
1 library(sf)
2
3 central_park <- st_read(file.path(
4   'data', 'central_park', 'CentralPark.shp'))
```

```
#> Reading layer `CentralPark' from data source `/Users/jhelvy/gh/t
#> Simple feature collection with 2550 features and 6 fields
#> Geometry type: LINESTRING
#> Dimension:     XY
#> Bounding box:  xmin: -73.99249 ymin: 40.7625 xmax: -73.93922 yma
#> Geodetic CRS:  WGS 84
```

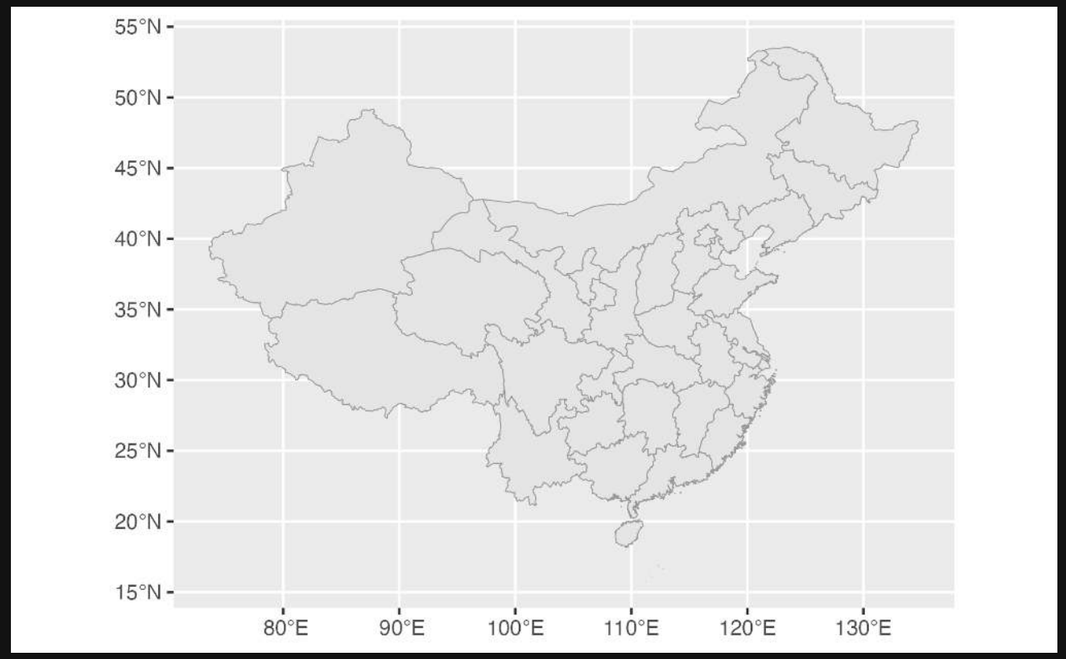
```
1 ggplot(data = central_park) +
2   geom_sf(color = 'grey75')
```

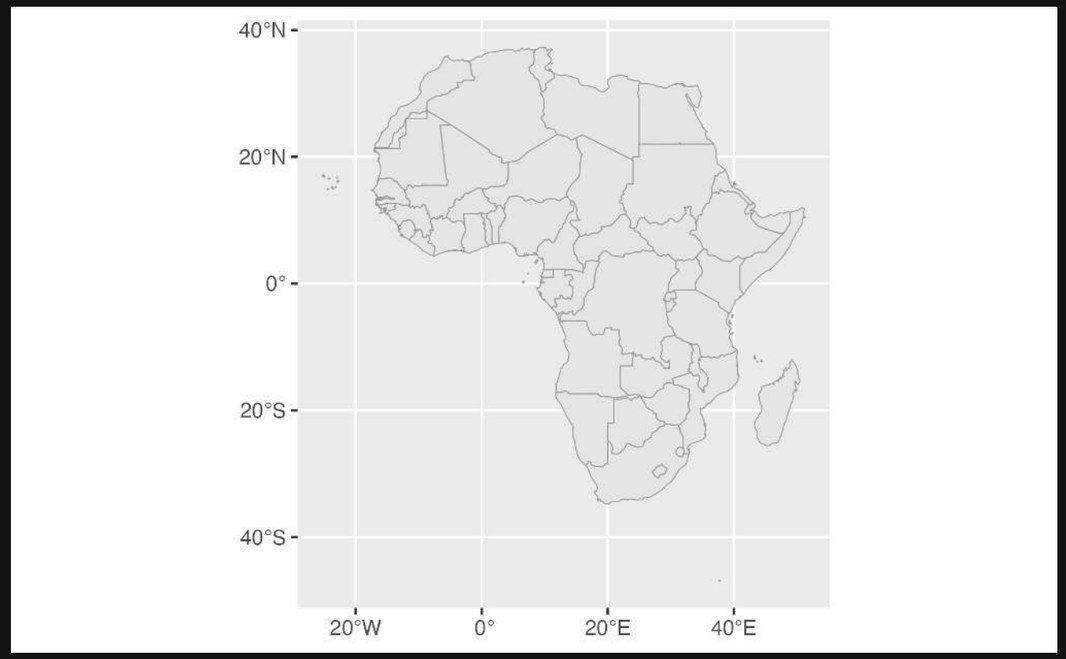


Your turn

10:00

Use the **rnaturalearth** library to extract and plot the shape files for China and Africa:



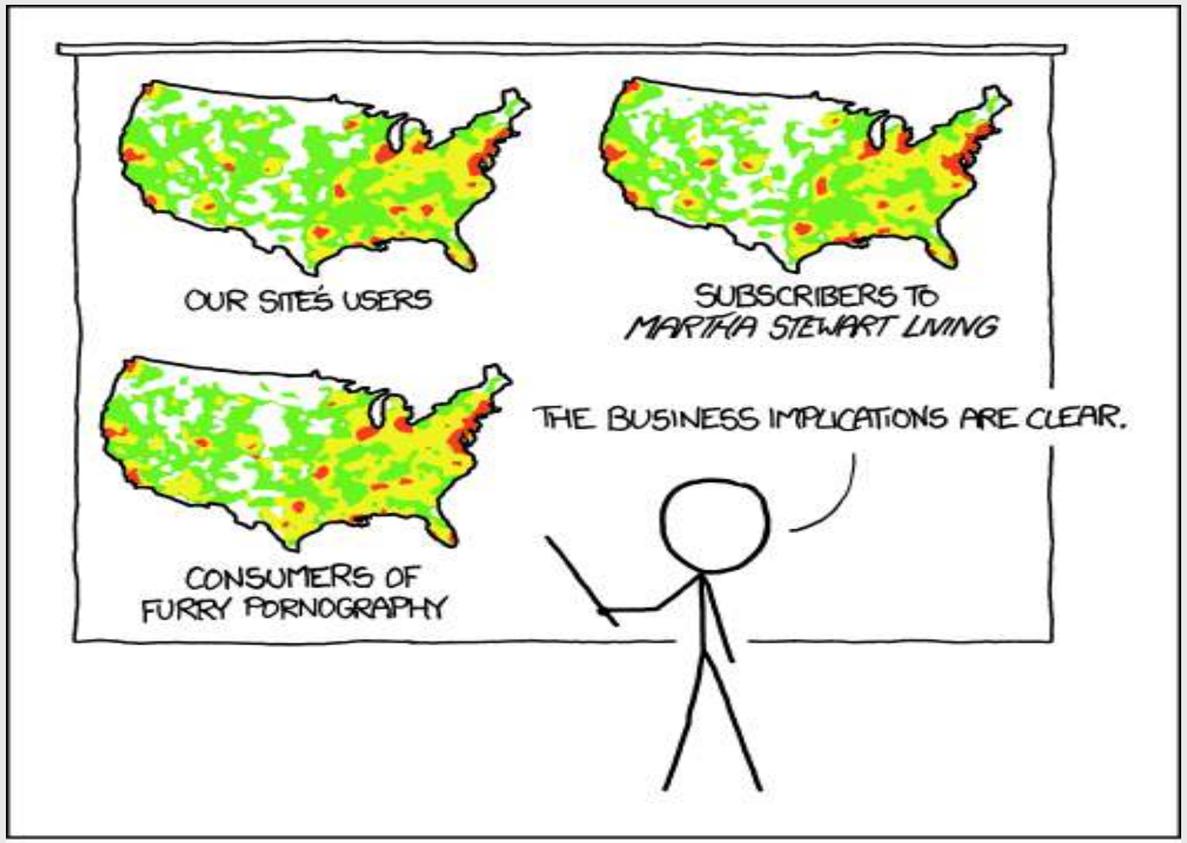


Week 12: *Maps*

1. Plotting maps
2. Adding data to maps
3. Projections

First rule of adding data to maps:

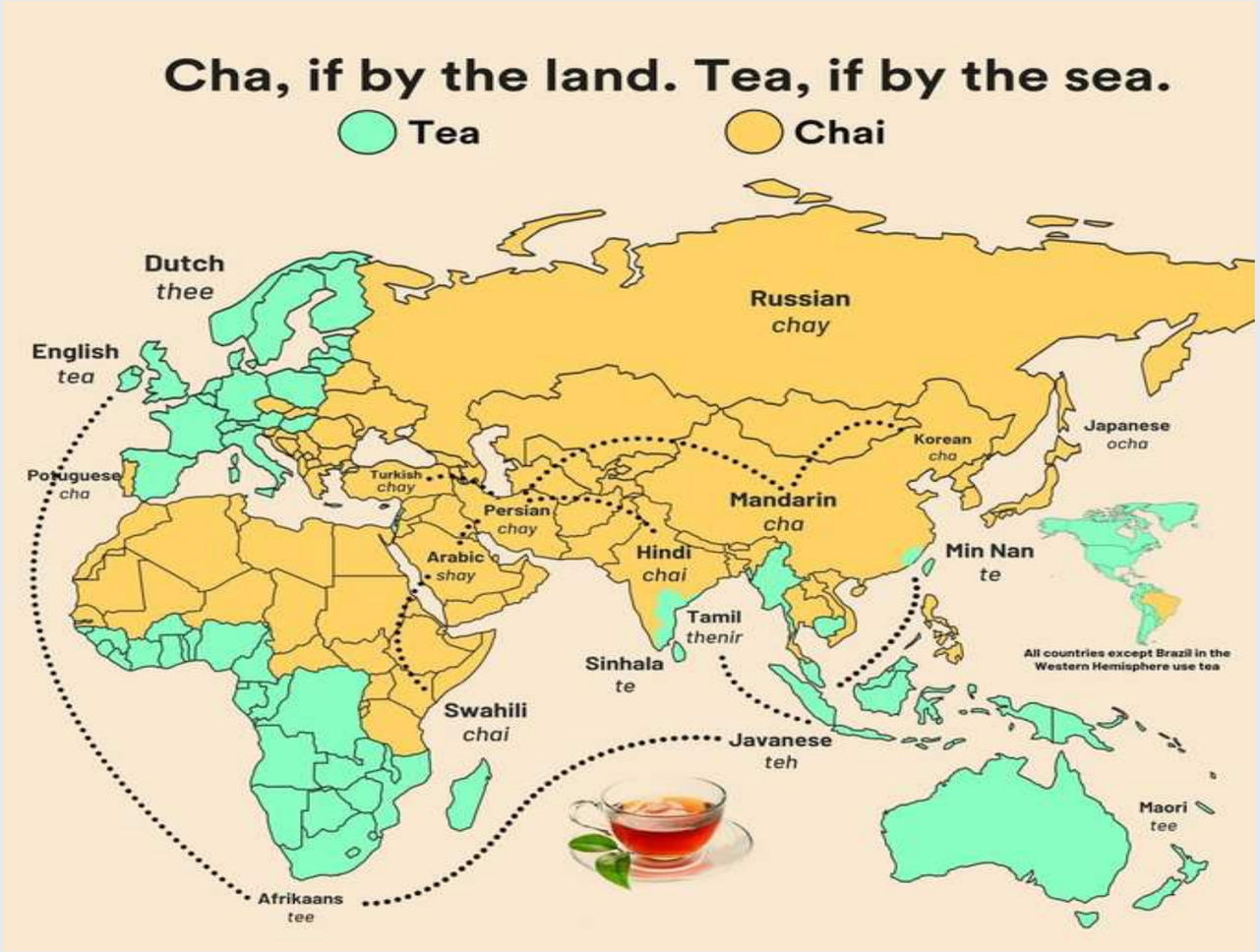
Do you need to make a map?



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE BASICALLY JUST POPULATION MAPS

https://xkcd.com/1138/

Not all maps are useful...



...but some maps are

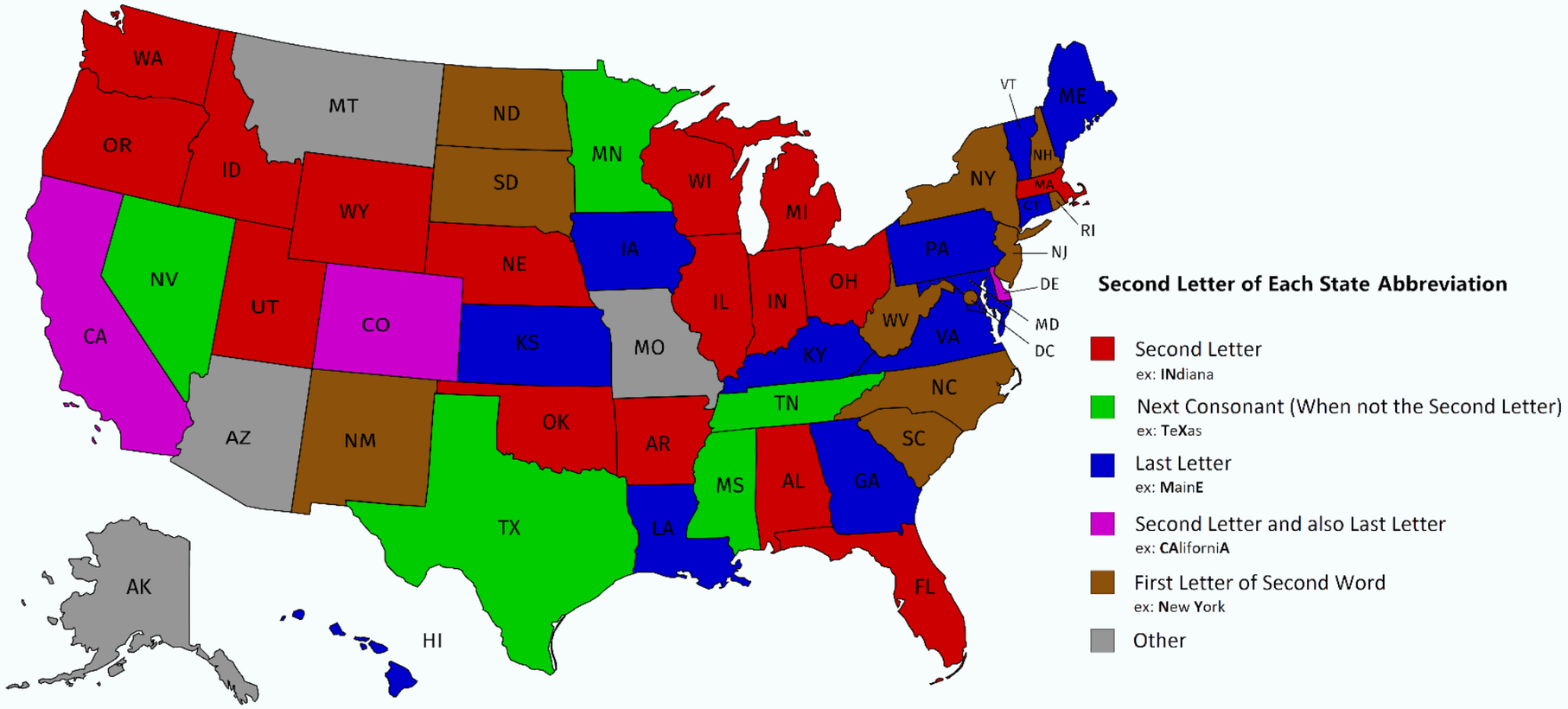
https://twitter.com/pwang/status/1387228573896003585

1. Choropleth maps

2. Point maps

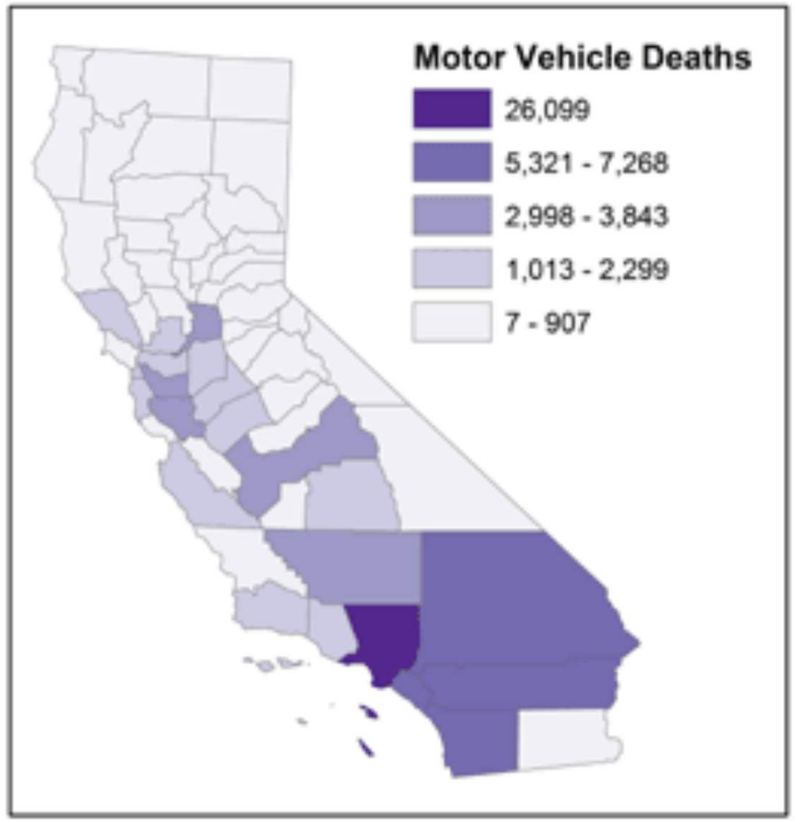
Choropleth - from Greek:

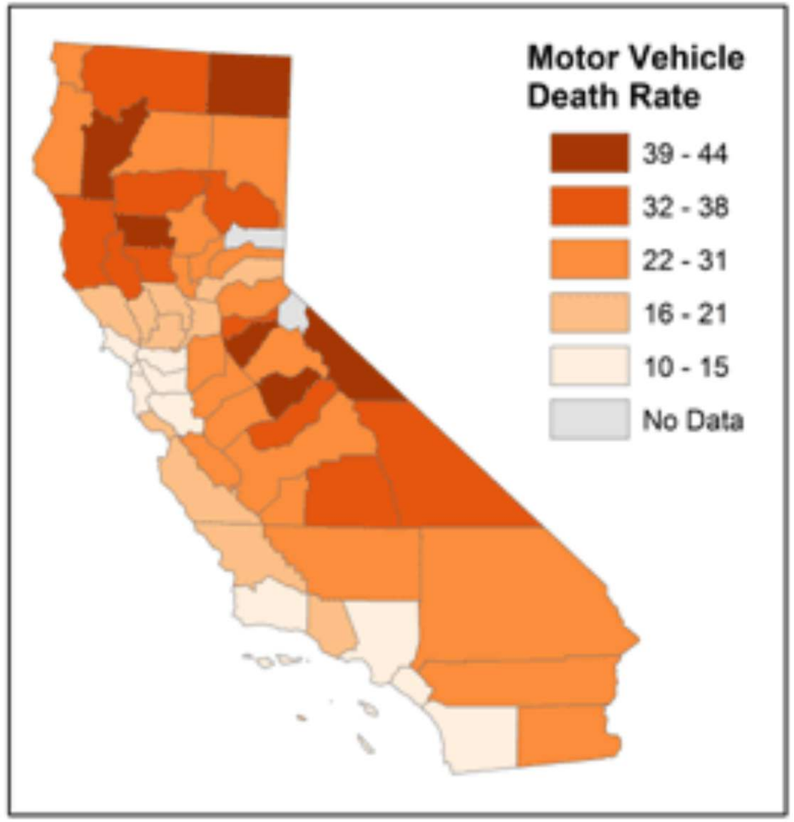
- χῶρος “choros” (area/region)
- πλῆθος “plethos” (multitude)



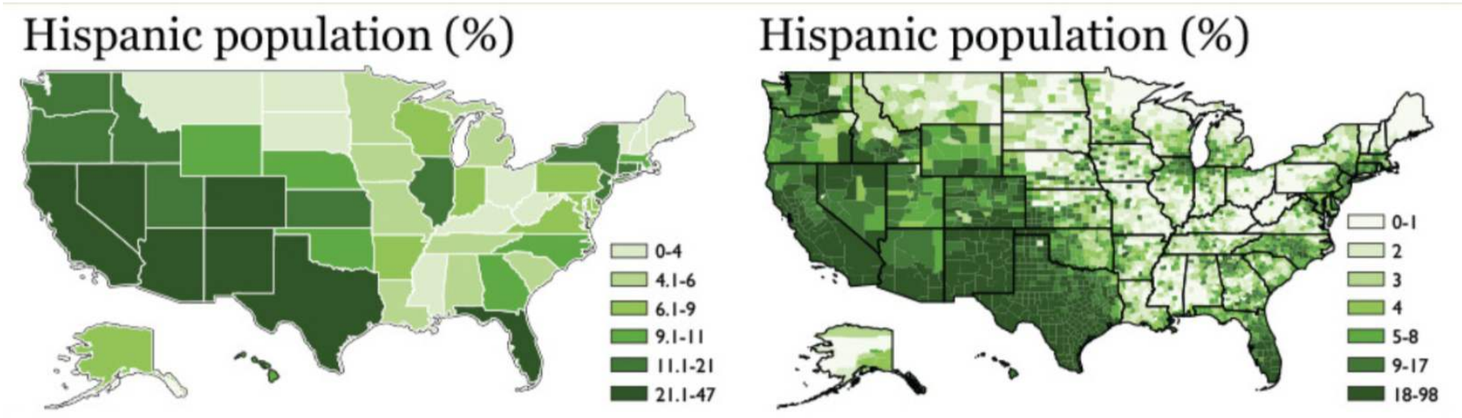
Choropleth maps are easily misleading

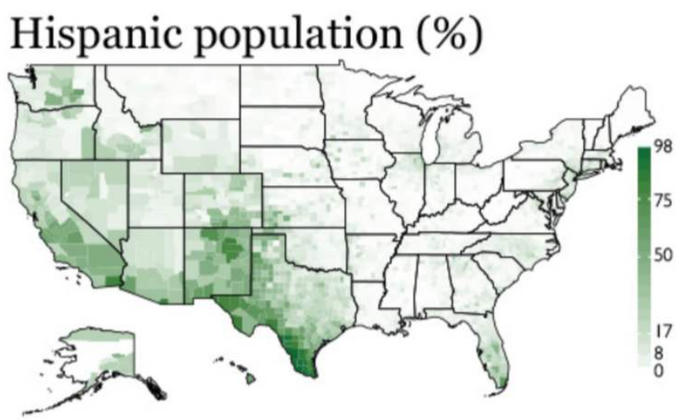
Number of events != Number of events **per capita**



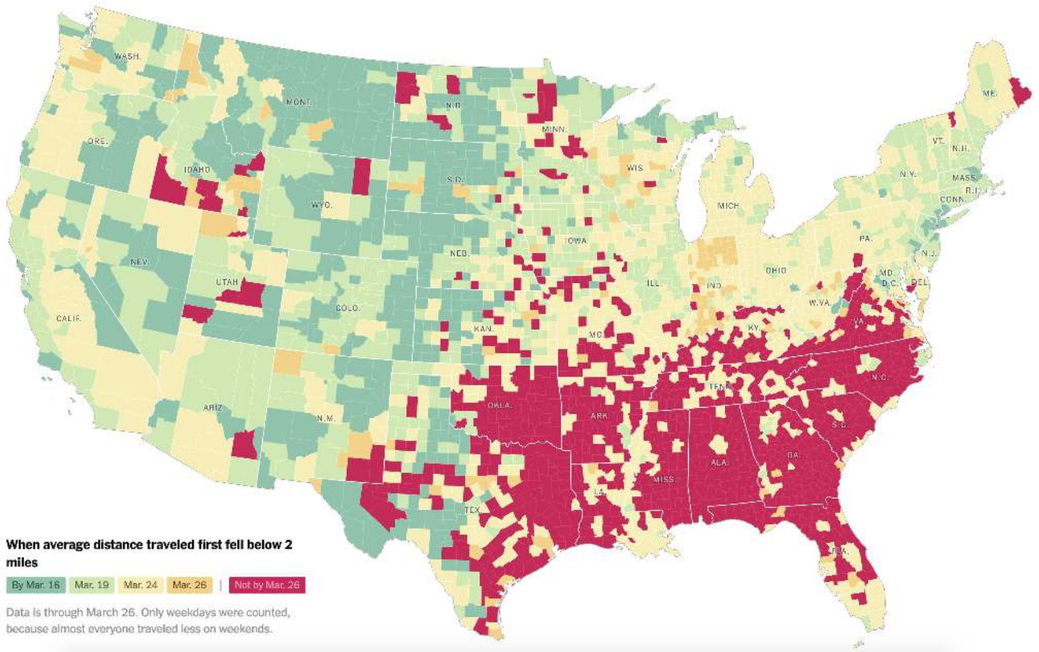


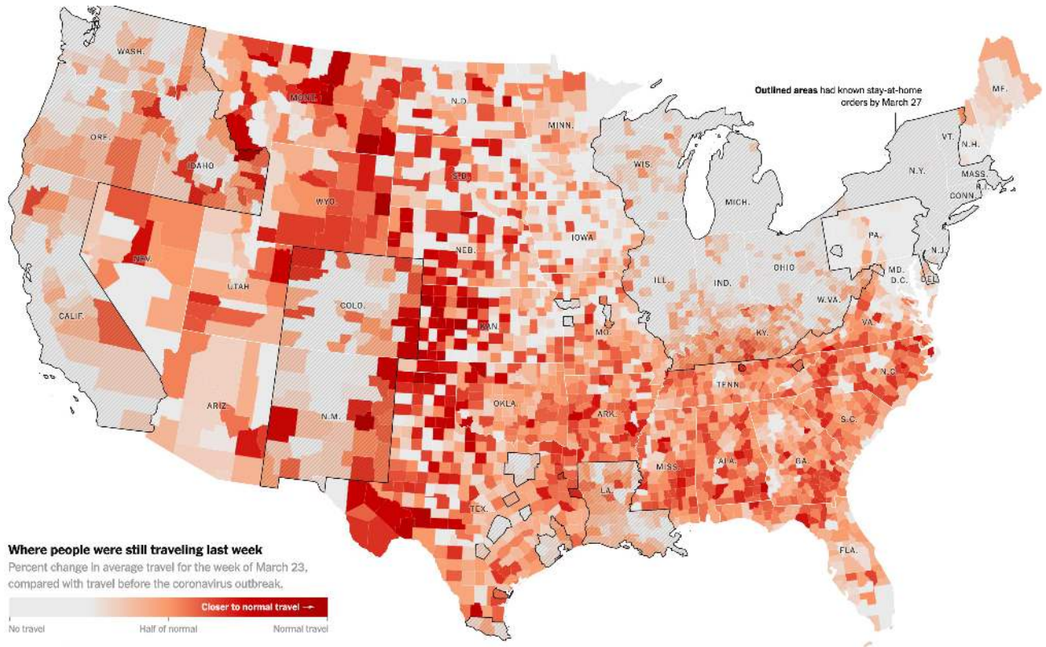
Manipulating fill scale produces wildly different maps





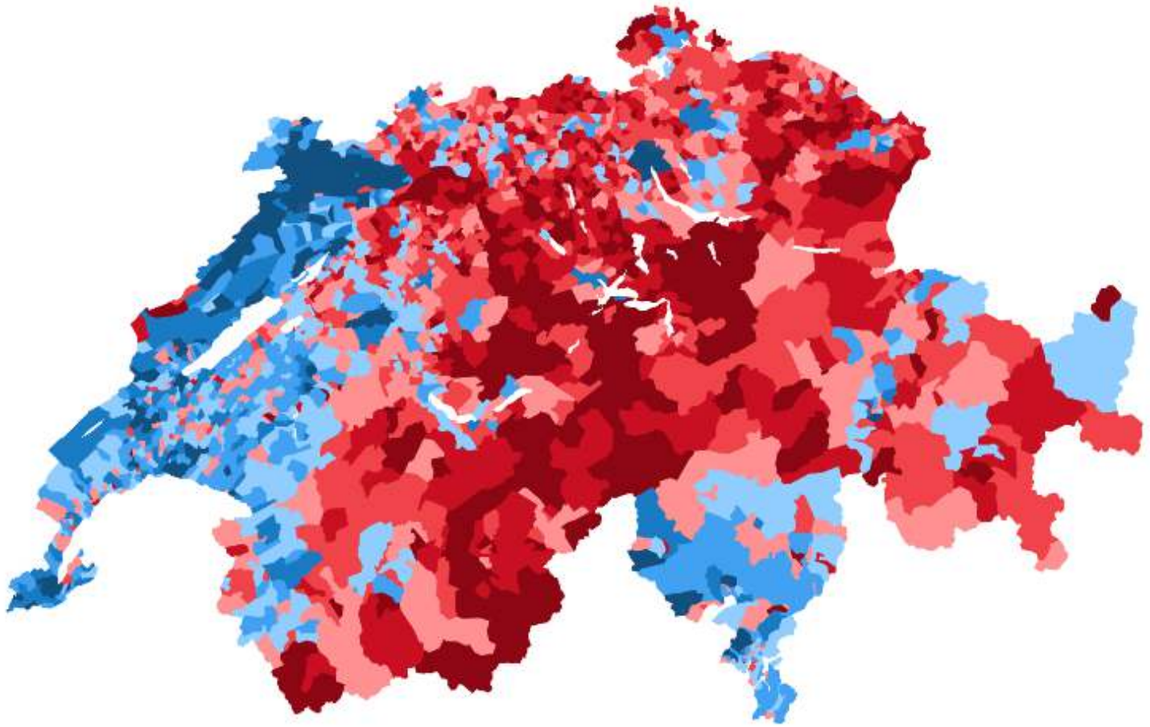
Manipulating fill scale produces wildly different maps





Source: New York Times

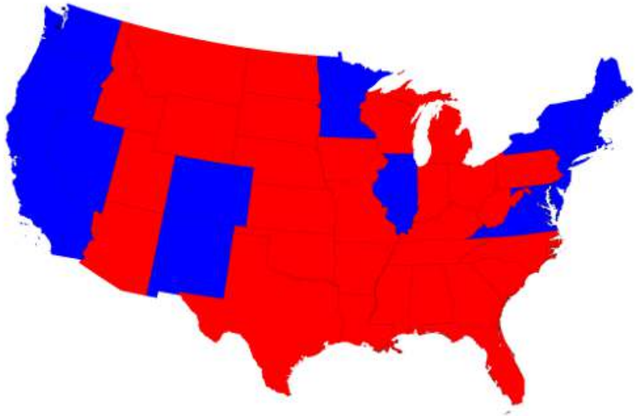
Land doesn’t vote - people vote

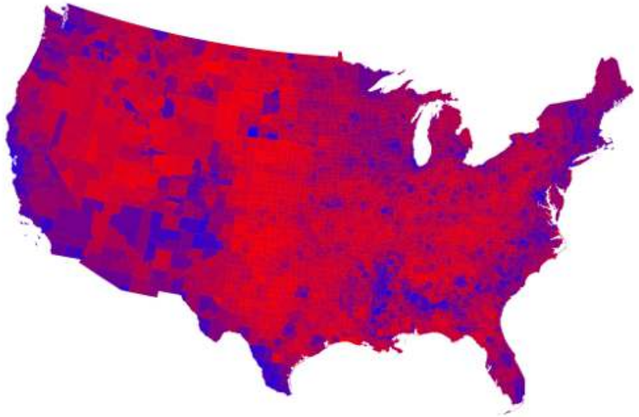


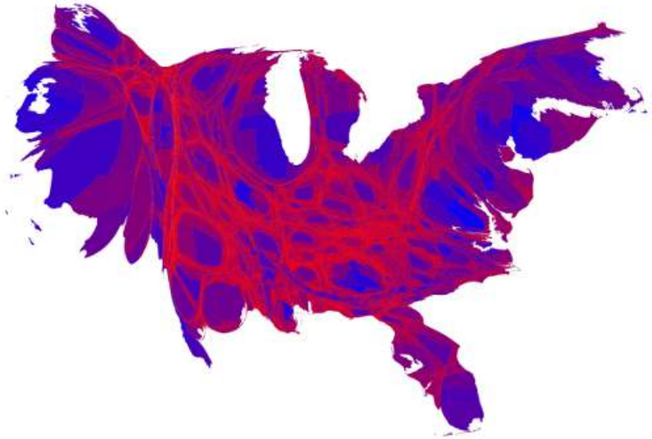
By David Zumbach

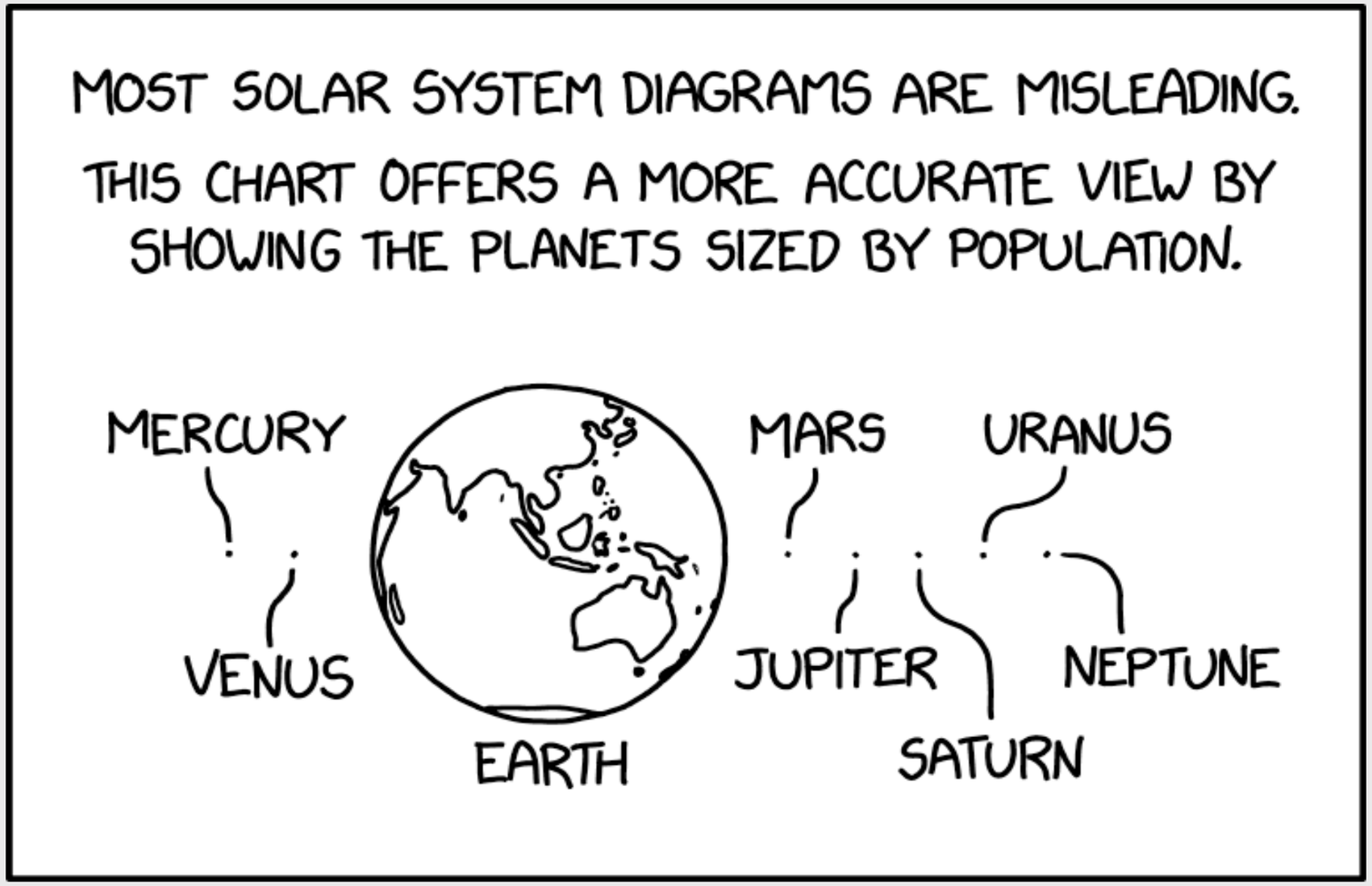
Land doesn’t vote - people vote

Election maps from: http://www-personal.umich.edu/~mejn/election/2016/









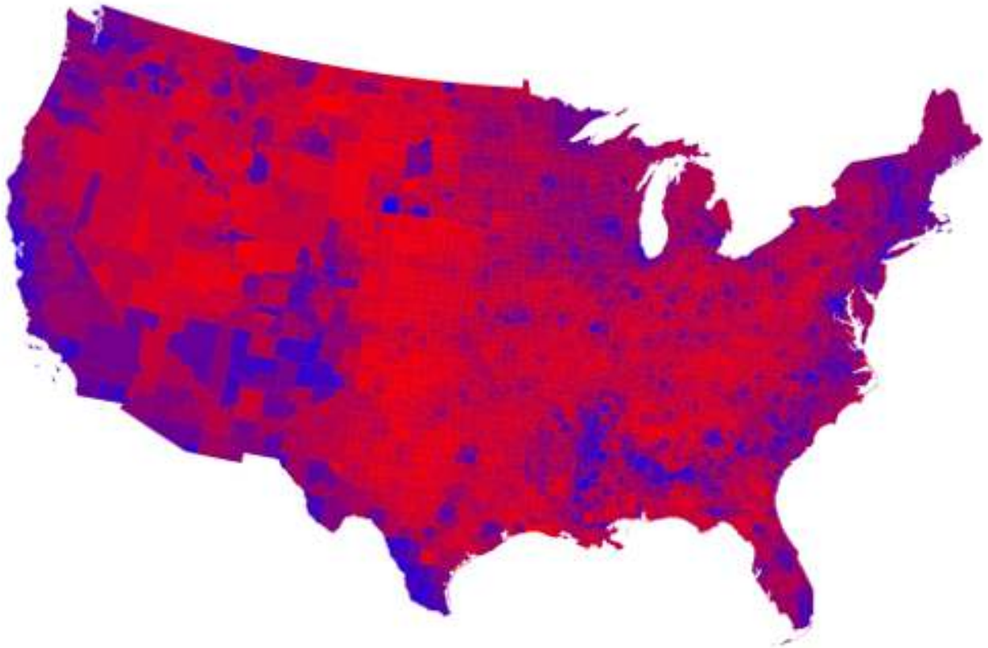
https://xkcd.com/2439/

Easy to lie with fake news

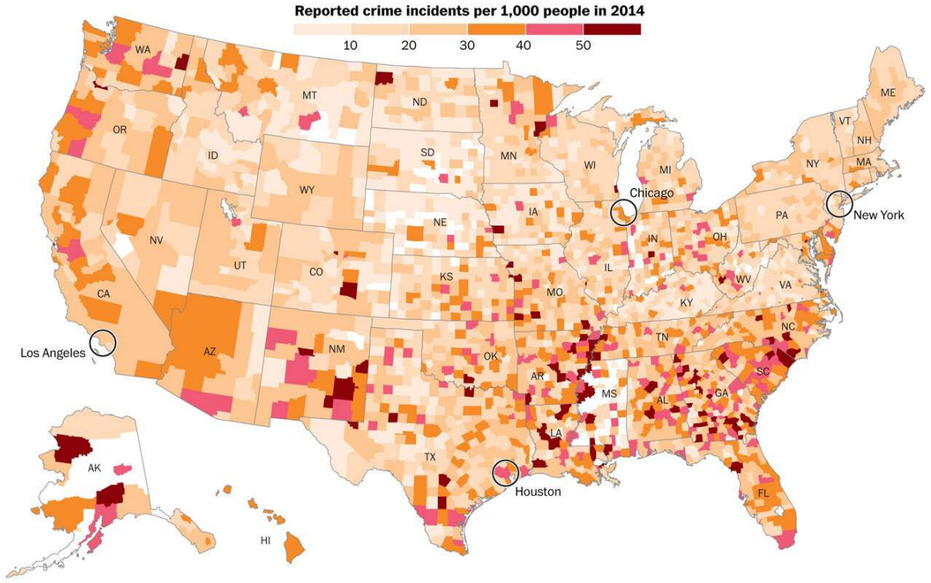


(here is what actual crime rates look like)

2016 Election map [source]

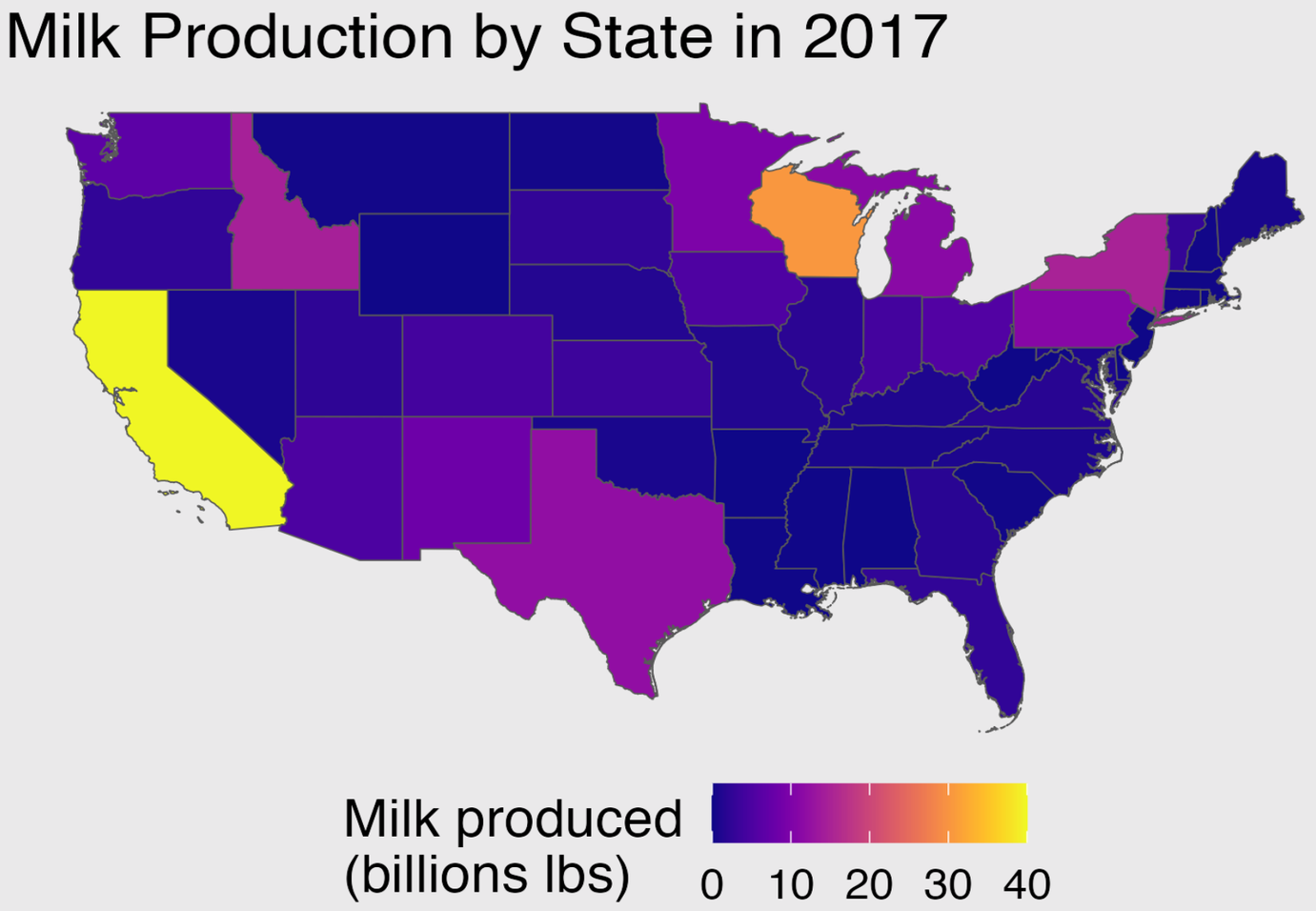


2014 Crime map [source]



A choropleth alternative: hex maps

How to make a choropleth map



How to make a choropleth map

Get the “fill” data

```
milk_2017 <- milk_production %>%
    filter(year == 2017) %>%
    select(name = state, milk_produced) %>%
    mutate(milk_produced = milk_produced / 10^9)
```

Get the “map” data

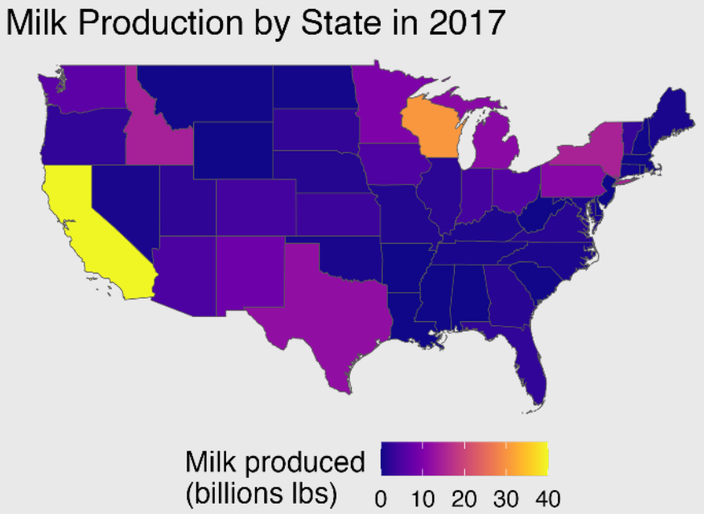
```
us_states <- ne_states(
    country = 'united states of america',
    returnclass = 'sf') %>%
    filter(! name %in% c('Alaska', 'Hawaii')) %>%
    left_join(milk_2017, by = 'name')
```

```
us_states %>%
    select(name, milk_produced) %>%
    head()
```

```
#> Simple feature collection with 6 fea
#> Geometry type: MULTIPOLYGON
#> Dimension:     XY
#> Bounding box:  xmin: -124.7346 ymin:
#> Geodetic CRS:  WGS 84
#>           name milk_produced
#> 1   Washington         6.526 MULTIPO
#> 2        Idaho        14.627 MULTIPO
#> 3      Montana         0.288 MULTIPO
#> 4 North Dakota         0.345 MULTIPO
#> 5    Minnesota         9.864 MULTIPO
#> 6     Michigan        11.231 MULTIPO
```

How to make a choropleth map

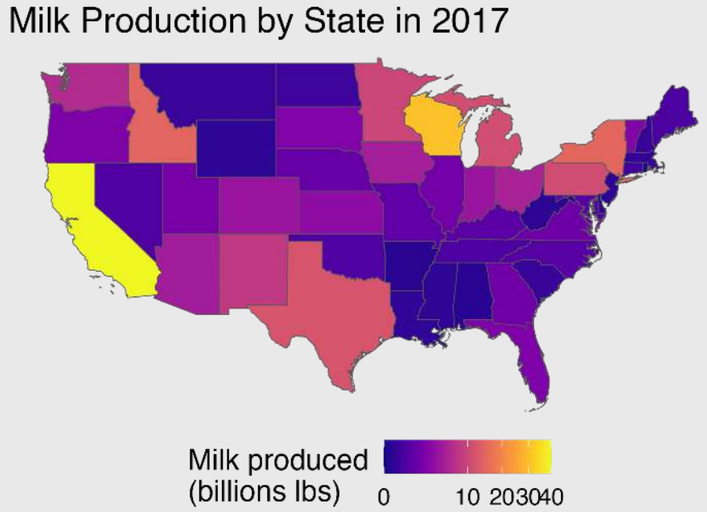
```
ggplot(us_states) +
  geom_sf(aes(fill = milk_produced)) +
  scale_fill_viridis(
    option = "plasma",
    limits = c(0, 40)) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  labs(
    fill = 'Milk produced\n(billions lbs)',
    title = 'Milk Production by State in 2017'
  )
```



How to make a choropleth map

```
ggplot(us_states) +
  geom_sf(aes(fill = milk_produced)) +
  scale_fill_viridis(
    trans = 'sqrt',
    option = "plasma",
    limits = c(0, 40)) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  labs(
    fill = 'Milk produced\n(billions lbs)',
    title = 'Milk Production by State in 2017'
  )
```

Non-linear scale:



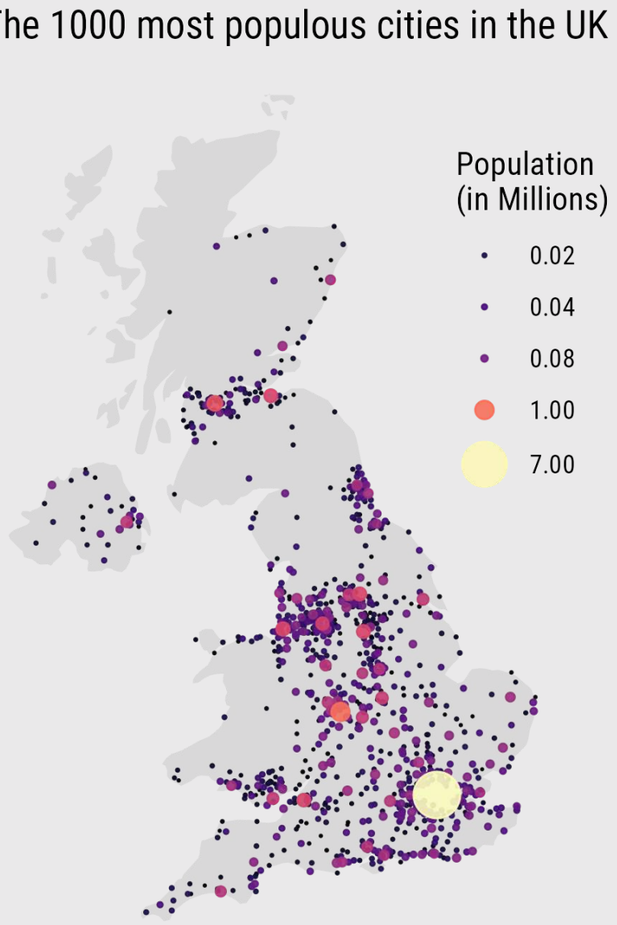
1. Choropleth maps

2. Point maps

Points as locations



Points encoding a variable

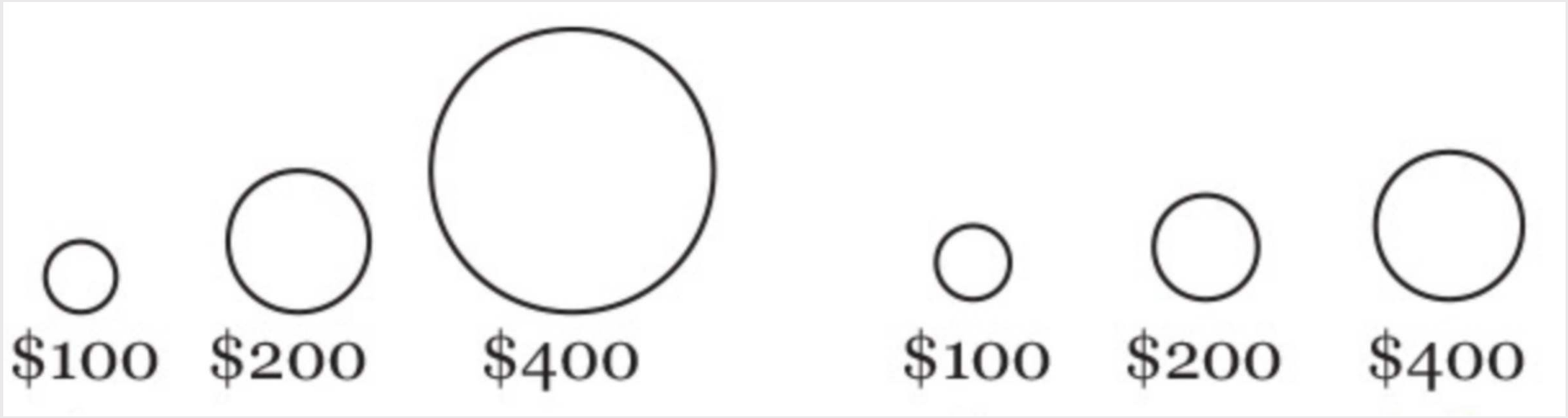


For point size, use **area**, not radius

$$Area = \pi r^2$$

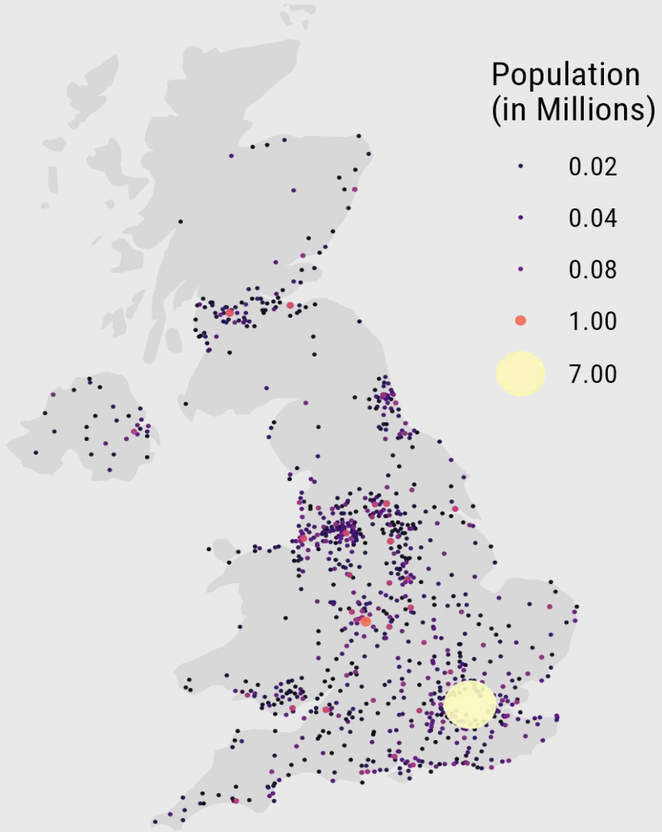
Radius

Area



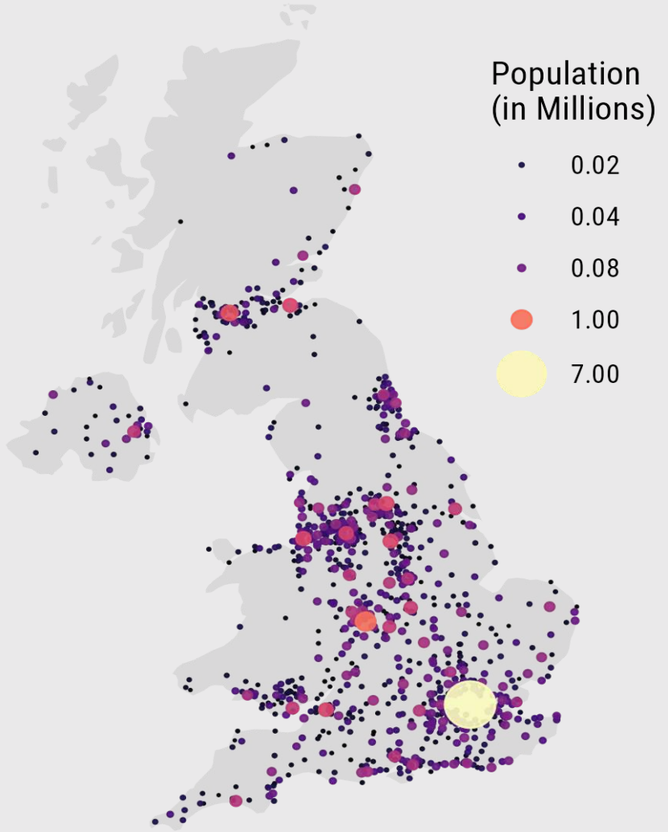
Radius

The 1000 most populous cities in the UK

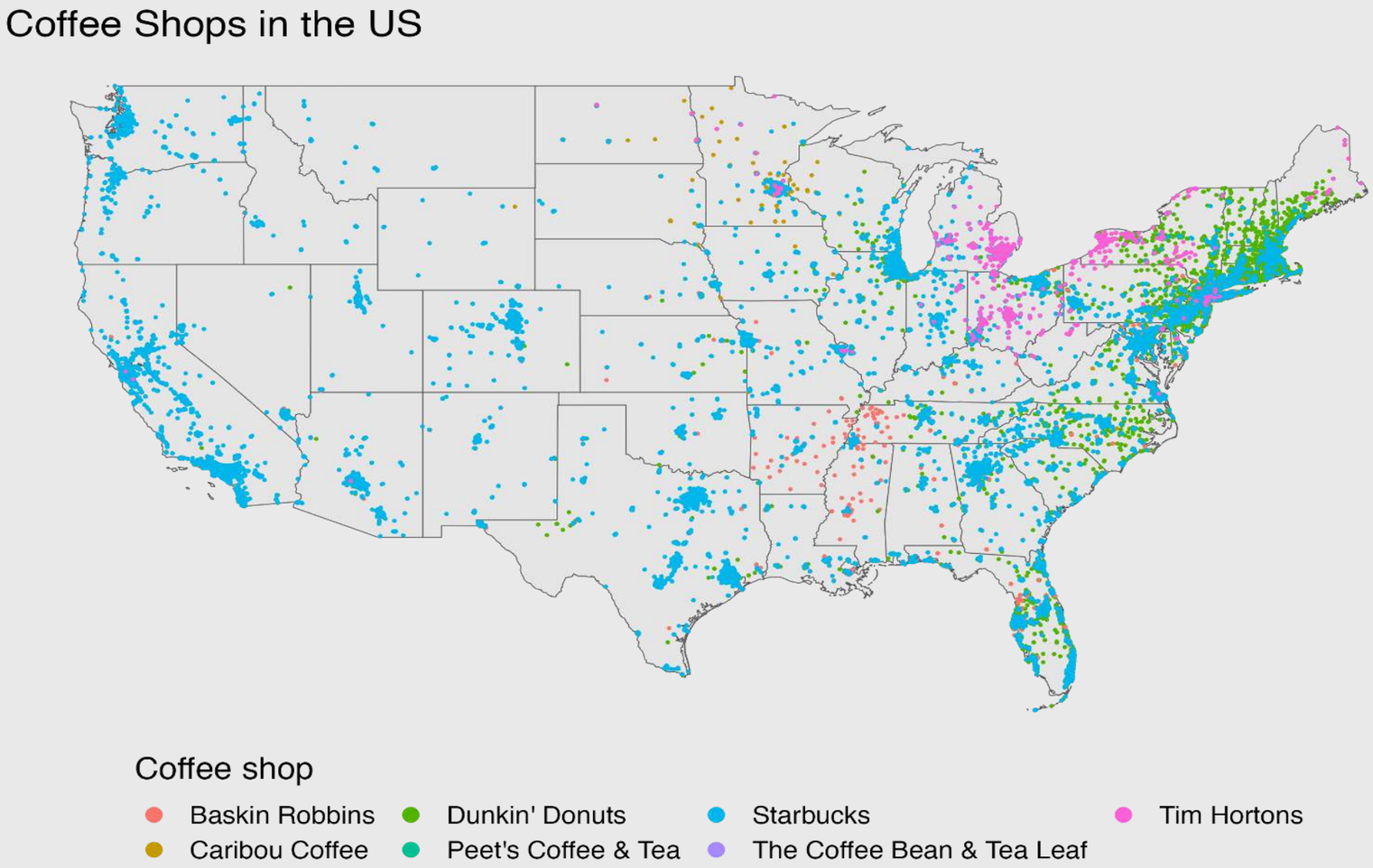


Area

The 1000 most populous cities in the UK



How to add points to a map



How to add points to a map

Load the continental US shape file

```
us_states_cont <- ne_states(
    country = 'united states of america',
    returnclass = 'sf') %>%
    filter(! name %in% c('Alaska', 'Hawaii'))
```

Read in the coffee shop data

```
us_coffee_shops  <- read_csv(file.path(
    'data', 'us_coffee_shops.csv'))

# Only keep data in continental US
us_coffee_shops <- us_coffee_shops %>%
    filter(
      lat > 22,    lat < 50,
      long > -150, long < -66
    )
```

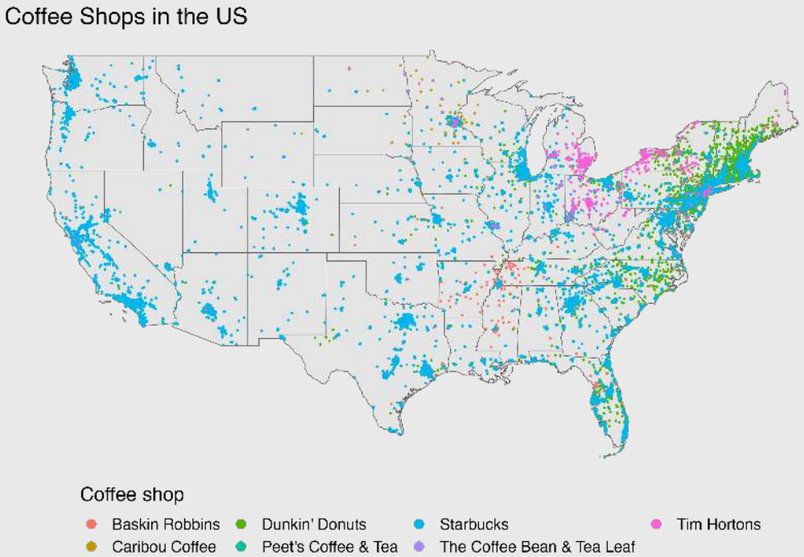
```
head(us_coffee_shops)
```

```
#> # A tibble: 6 × 8
#>   name              lat  long unique_id city
#>   <chr>           <dbl> <dbl>     <dbl> <chr>
#> 1 Baskin Robbins  40.8 -73.4   3304448 Hunting
#> 2 Baskin Robbins  42.1 -88.0  11342048 Rolling
#> 3 Baskin Robbins  34.0 -84.5   3304169 Mariett
#> 4 Baskin Robbins  29.8 -95.6   3304006 Houston
#> 5 Baskin Robbins  36.4 -89.5   3303959 Tiptonv
#> 6 Baskin Robbins  40.7 -73.6   3304507 Merrick
```

How to add points to a map

Plot coffee shop locations over map

```
ggplot() +
  geom_sf(data = us_states_cont) +
  geom_point(
    data = us_coffee_shops,
    aes(x = long, y = lat, color = name),
    size = 0.3
  ) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  guides(color = guide_legend(
    # Move legend title to top
    title.position = "top",
    # Increase legend point size
    override.aes = list(size = 3))) +
  labs(
    color = 'Coffee shop',
    title = 'Coffee Shops in the US'
  )
```



Your turn

15:00

Create this map of squirrels in NYC's Central Park using this data from the Squirrel Census:

- The `CentralPark.shp` file in the `data/central_park` folder.
- The `nyc_squirrels.csv` file in the `data` folder.

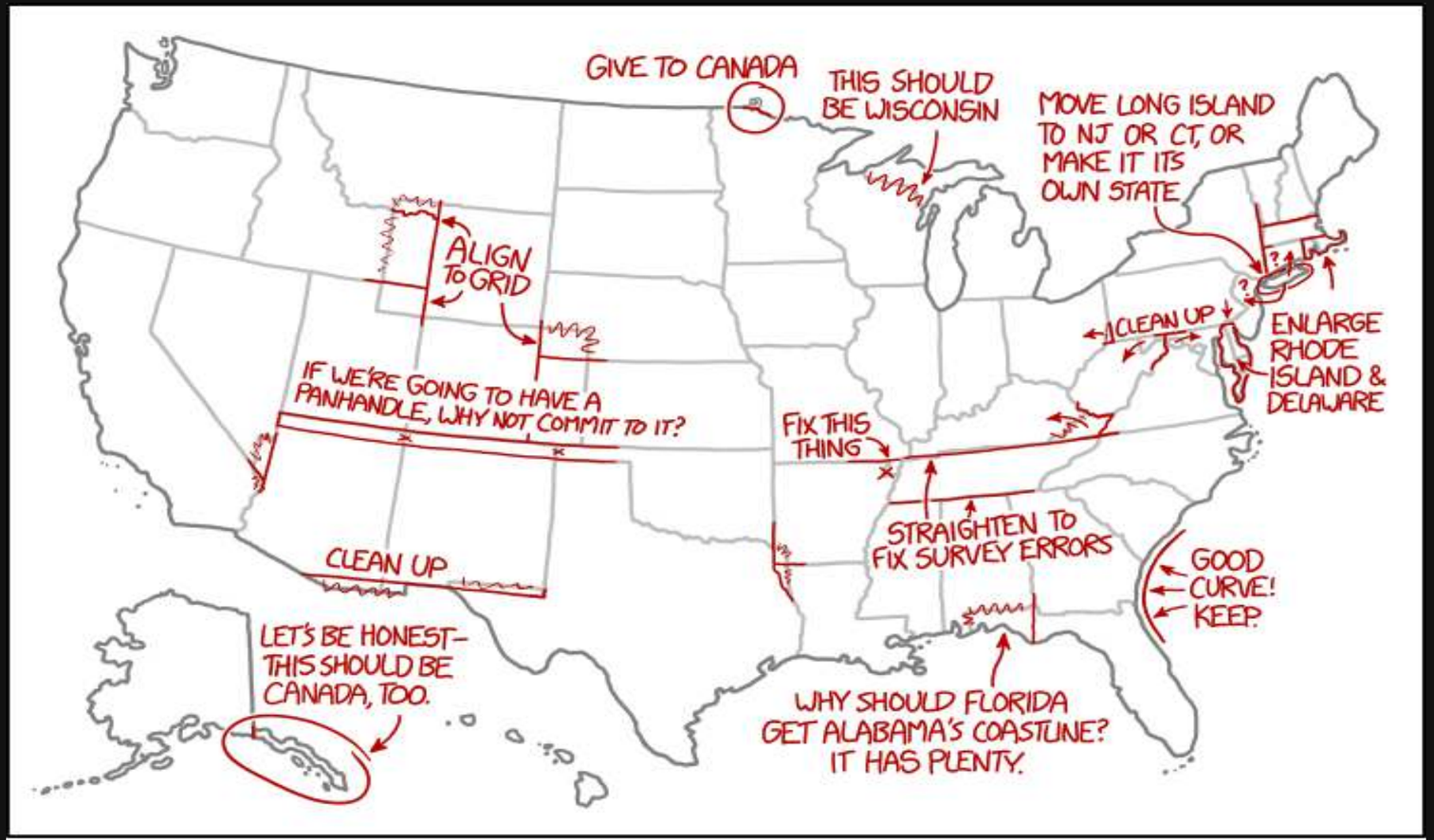
Hint: The color is mapped to the `primary_fur_color` variable

(More about the Squirrel Census here)



Intermission

05:00

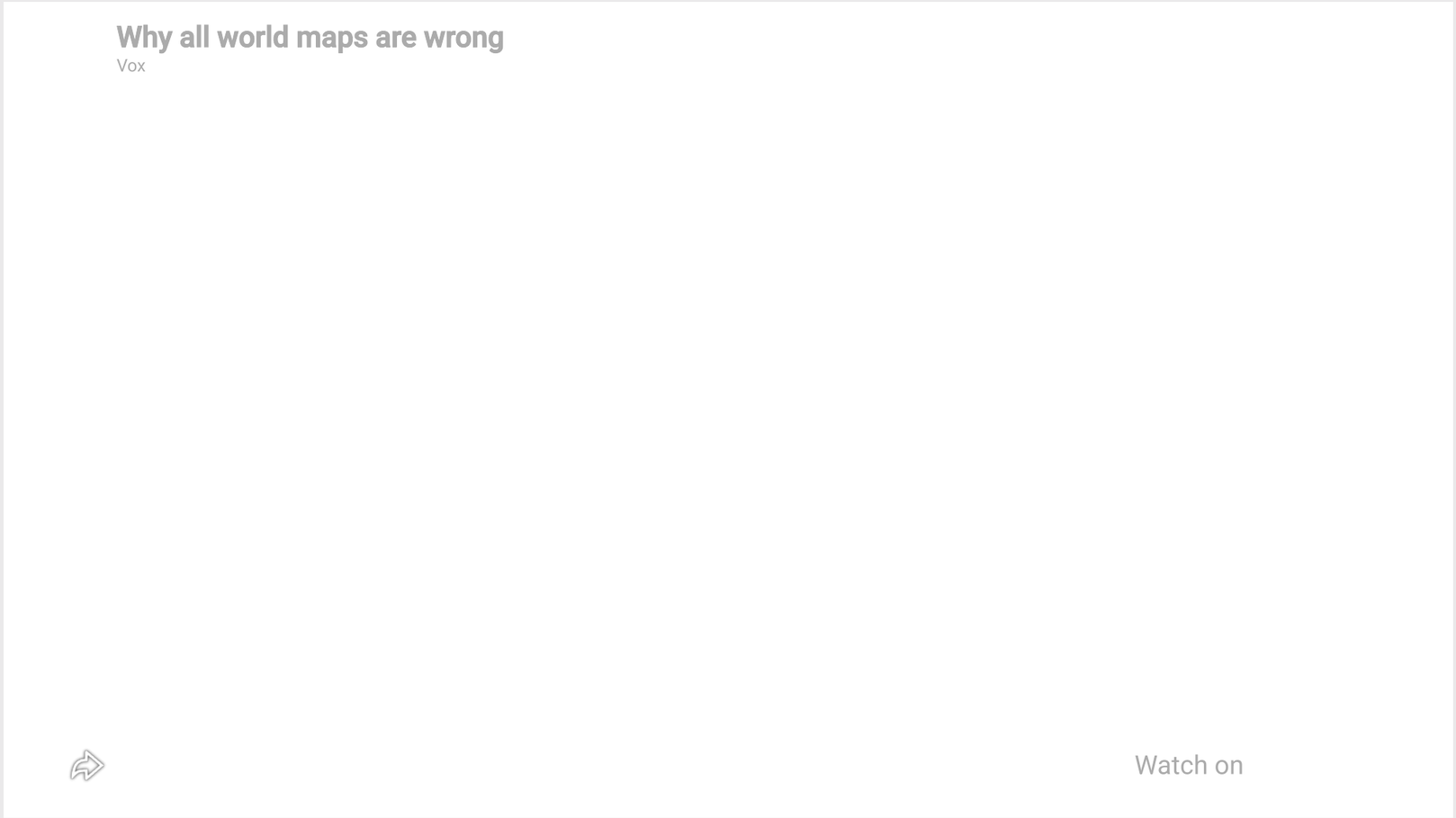


IT WAS SCARY WHEN THE GRAPHIC DESIGNERS SEIZED CONTROL OF THE COUNTRY, BUT IT TURNED OUT THEY JUST WANTED TO FIX SOME THINGS ABOUT THE STATE BORDERS THAT HAD ALWAYS BOTHERED THEM.

Week 12: *Maps*

1. Plotting maps
2. Adding data to maps
3. Projections

What’s a map projection?



What is the best projection?...it depends

Compare country sizes

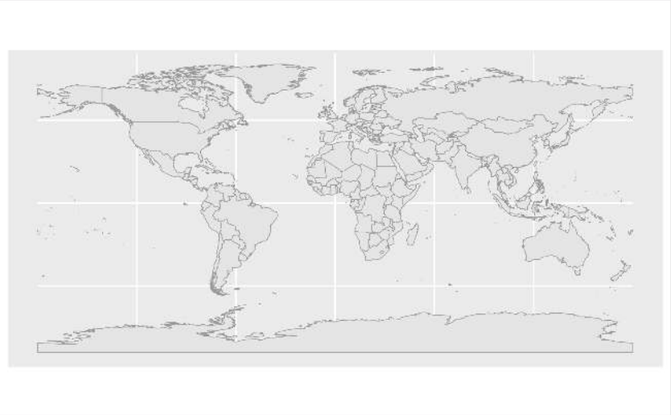
Using projections

To modify the projection of a map, use `coord_sf(crs = st_crs(XXXX))`

```
1  world <- ne_countries(scale = "medium", returnclass = "sf")
```

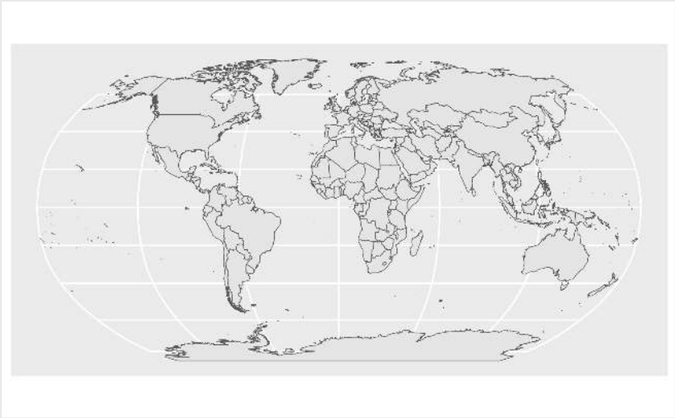
Default (long-lat)

```
1  ggplot(data = world) +
2      geom_sf()
```



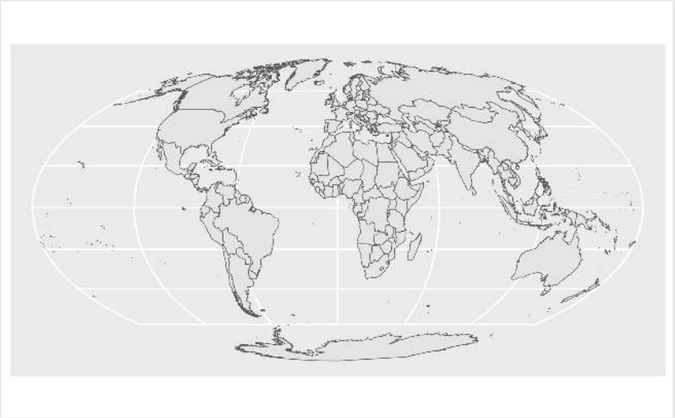
Robinson projection

```
1  ggplot(data = world) +
2      geom_sf() +
3      coord_sf(crs = "ESRI:54030")
```



Mollweide projection

```
1  ggplot(data = world) +
2      geom_sf() +
3      coord_sf(crs = "ESRI:54009")
```



Common Projections

ggplot layer:

```
1  coord_sf(crs = "ESRI:XXXX")
```

World

Code	Projection
"ESRI:54030"	**Robinson**
"ESRI:54002"	Equidistant cylindrical
"ESRI:54004"	Mercator
"ESRI:54008"	Sinusoidal
"ESRI:54009"	Mollweide

United States

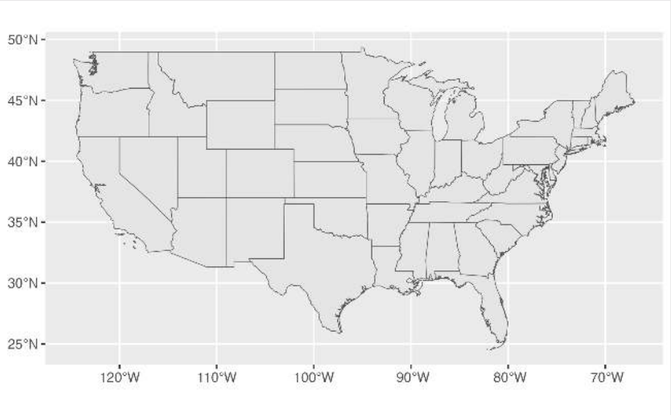
Code	Projection
"ESRI:102003"	**Albers**
"ESRI:102004"	Lambert Conformal Conic
4269	NAD 83

US projections

```
us_states_cont <- ne_states(country = 'united states of america',
    returnclass = 'sf') %>%
    filter(! name %in% c('Alaska', 'Hawaii'))
```

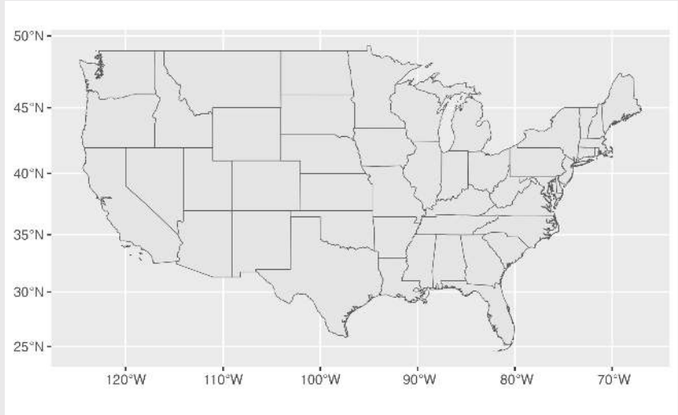
NAD 83 projection

```
ggplot(data = world) +
    geom_sf() +
    coord_sf(crs = 4269)
```



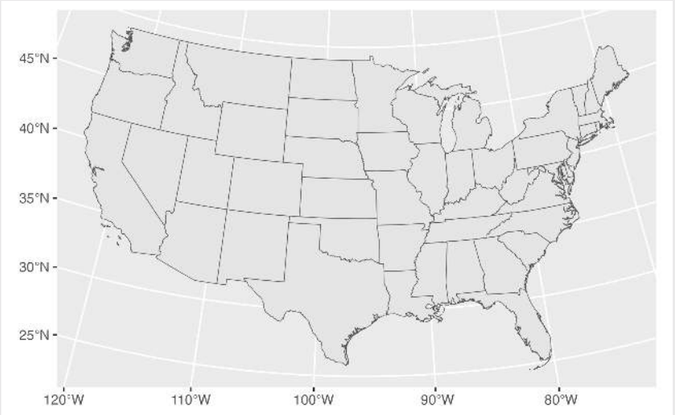
Mercator

```
ggplot(data = world) +
    geom_sf() +
    coord_sf(crs = "ESRI:54004")
```



Albers

```
ggplot(data = us_states_cont) +
    geom_sf() +
    coord_sf(crs = "ESRI:102003")
```

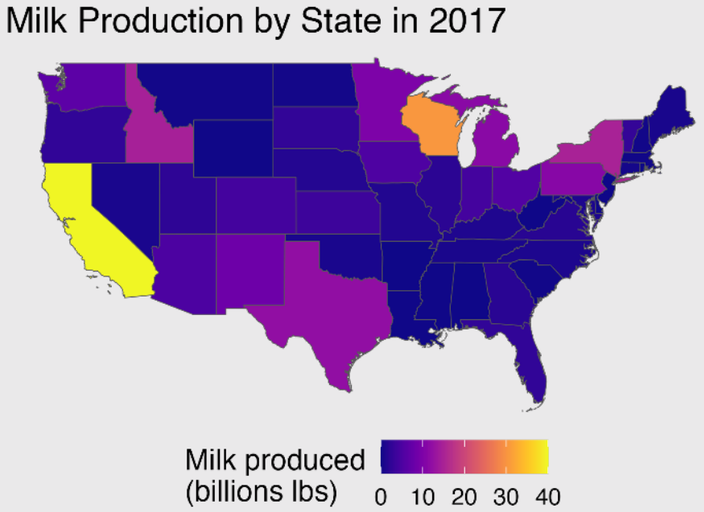


Mapping data to projections - choropleth map

```r
milk_2017 <- milk_production %>%
  filter(year == 2017) %>%
  select(name = state, milk_produced) %>%
  mutate(milk_produced = milk_produced / 10^9)

us_states <- ne_states(
  country = 'united states of america',
  returnclass = 'sf') %>%
  filter(! name %in% c('Alaska', 'Hawaii')) %>%
  left_join(milk_2017, by = 'name')

ggplot(us_states) +
  geom_sf(aes(fill = milk_produced)) +
  scale_fill_viridis(
    option = "plasma",
    limits = c(0, 40)) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  labs(
    fill = 'Milk produced\n(billions lbs)',
    title = 'Milk Production by State in 2017'
  )
```



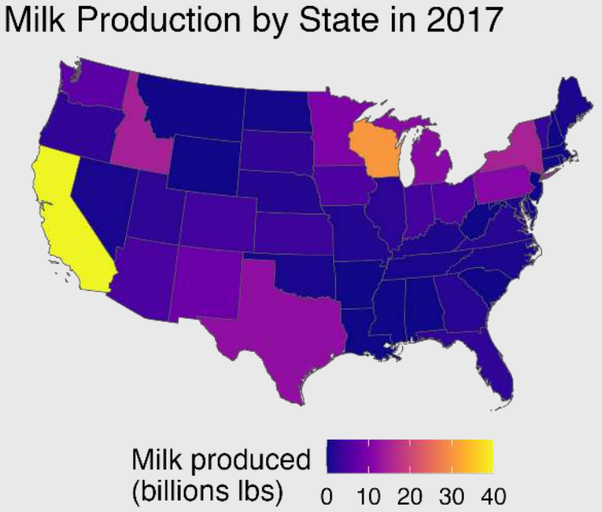
Mapping data to projections - choropleth map

```r
milk_2017 <- milk_production %>%
  filter(year == 2017) %>%
  select(name = state, milk_produced) %>%
  mutate(milk_produced = milk_produced / 10^9)

us_states <- ne_states(
  country = 'united states of america',
  returnclass = 'sf') %>%
  filter(! name %in% c('Alaska', 'Hawaii')) %>%
  left_join(milk_2017, by = 'name')

ggplot(us_states) +
  geom_sf(aes(fill = milk_produced)) +
  scale_fill_viridis(
    option = "plasma",
    limits = c(0, 40)) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  labs(
    fill = 'Milk produced\n(billions lbs)',
    title = 'Milk Production by State in 2017'
  ) +
  coord_sf(crs = "ESRI:102003")
```

Albers Projection

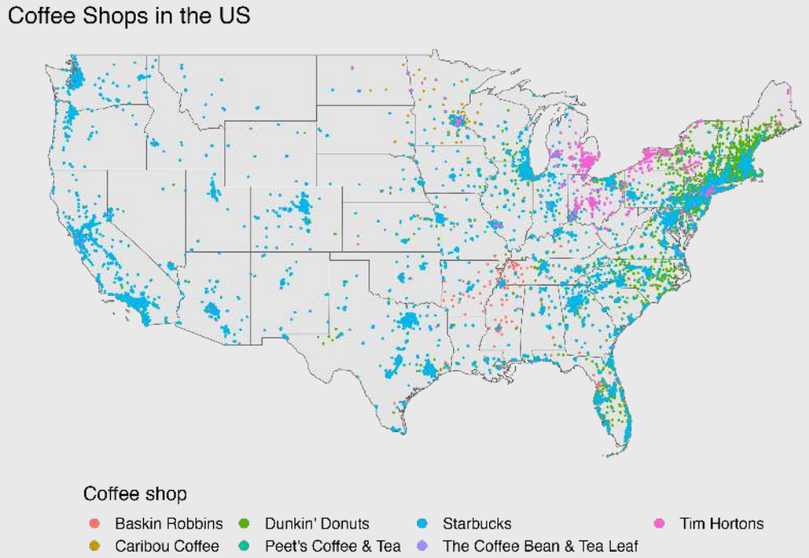


Mapping data to projections - points

```r
us_states_cont <- ne_states(
  country = 'united states of america',
  returnclass = 'sf') %>%
  filter(! name %in% c('Alaska', 'Hawaii'))

us_coffee_shops <- us_coffee_shops %>%
  filter(lat > 22,    lat < 50,
         long > -150, long < -66)

ggplot() +
  geom_sf(data = us_states_cont) +
  geom_point(
    data = us_coffee_shops,
    aes(x = long, y = lat, color = name),
    size = 0.3) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  guides(color = guide_legend(
    # Move legend title to top
    title.position = "top",
    # Increase legend point size
    override.aes = list(size = 3))) +
  labs(
    color = 'Coffee shop',
    title = 'Coffee Shops in the US'
  )
```



Mapping data to projections - points

```
us_states_cont <- ne_states(
  country = 'united states of america',
  returnclass = 'sf') %>%
  filter(! name %in% c('Alaska', 'Hawaii'))

us_coffee_shops <- us_coffee_shops %>%
  filter(lat > 22,    lat < 50,
         long > -150, long < -66)

ggplot() +
  geom_sf(data = us_states_cont) +
  geom_point(
    data = us_coffee_shops,
    aes(x = long, y = lat, color = name),
    size = 0.3) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  guides(color = guide_legend(
    # Move legend title to top
    title.position = "top",
    # Increase legend point size
    override.aes = list(size = 3))) +
  labs(
    color = 'Coffee shop',
    title = 'Coffee Shops in the US'
  ) +
  coord_sf(crs = "ESRI:102003")
```

Fail!



Mapping data to projections - points

First match `us_coffee_shops` crs to `us_states_cont`

```
us_states_cont <- ne_states(
    country = 'united states of america',
    returnclass = 'sf') %>%
    filter(! name %in% c('Alaska', 'Hawaii'))

us_coffee_shops <- us_coffee_shops %>%
    filter(lat > 22,    lat < 50,
           long > -150, long < -66)

us_coffee_shops_sf <- st_as_sf(us_coffee_shops,
    coords = c("long", "lat"),
    crs = st_crs(us_states_cont))
```

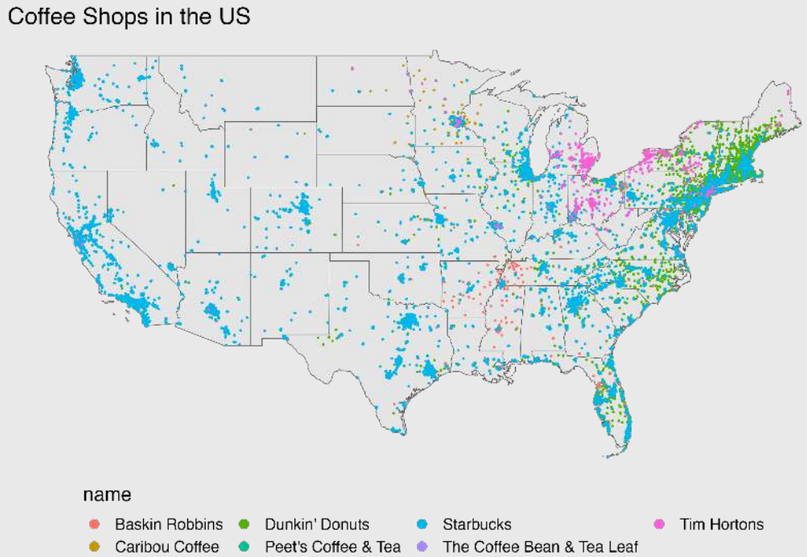
```
head(us_coffee_shops_sf)
```

```
#> Simple feature collection with 6 fea
#> Geometry type: POINT
#> Dimension:     XY
#> Bounding box:  xmin: -95.60337 ymin:
#> Geodetic CRS:  WGS 84
#> # A tibble: 6 × 7
#>   name           unique_id city
#>   <chr>              <dbl> <chr>
#> 1 Baskin Robbins   3304448 Huntingto
#> 2 Baskin Robbins  11342048 Rolling M
#> 3 Baskin Robbins   3304169 Marietta
#> 4 Baskin Robbins   3304006 Houston
#> 5 Baskin Robbins   3303959 Tiptonvil
#> 6 Baskin Robbins   3304507 Merrick
```

Mapping data to projections - points

Plot coffee shop locations over map with `geom_sf()`

```
ggplot() +
  geom_sf(data = us_states_cont) +
  geom_sf(
    data = us_coffee_shops_sf,
    aes(color = name),
    size = 0.3) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  guides(color = guide_legend(
    # Move legend title to top
    title.position = "top",
    # Increase legend point size
    override.aes = list(size = 3))) +
  labs(
    fill = 'Coffee shop',
    title = 'Coffee Shops in the US'
  )
```



Mapping data to projections - points

Plot coffee shop locations over map with `geom_sf()`

```
ggplot() +
  geom_sf(data = us_states_cont) +
  geom_sf(
    data = us_coffee_shops_sf,
    aes(color = name),
    size = 0.3) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  guides(color = guide_legend(
    # Move legend title to top
    title.position = "top",
    # Increase legend point size
    override.aes = list(size = 3))) +
  labs(
    fill = 'Coffee shop',
    title = 'Coffee Shops in the US'
  ) +
  coord_sf(crs = "ESRI:102003")
```

Albers Projection



Mapping data to projections - points

Plot coffee shop locations over map with `geom_sf()`

```
ggplot() +
  geom_sf(data = us_states_cont) +
  geom_sf(
    data = us_coffee_shops_sf,
    aes(color = name),
    size = 0.3) +
  theme_void(base_size = 15) +
  theme(legend.position = 'bottom') +
  guides(color = guide_legend(
    # Move legend title to top
    title.position = "top",
    # Increase legend point size
    override.aes = list(size = 3))) +
  labs(
    fill = 'Coffee shop',
    title = 'Coffee Shops in the US'
  ) +
  coord_sf(crs = "ESRI:102004")
```

LCC Projection



20:00

Your turn

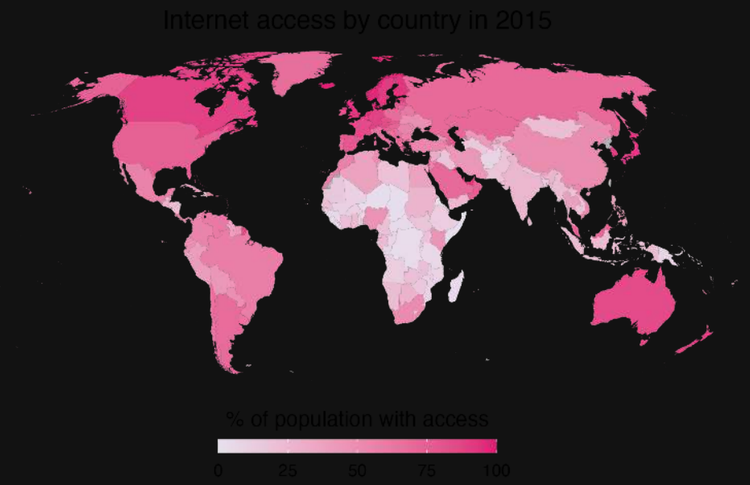
Use the `internet_users_country.csv` data and the `world` data frame from the **rnaturalearth** library to create these two versions of internet access by country in 2015.

Hints:

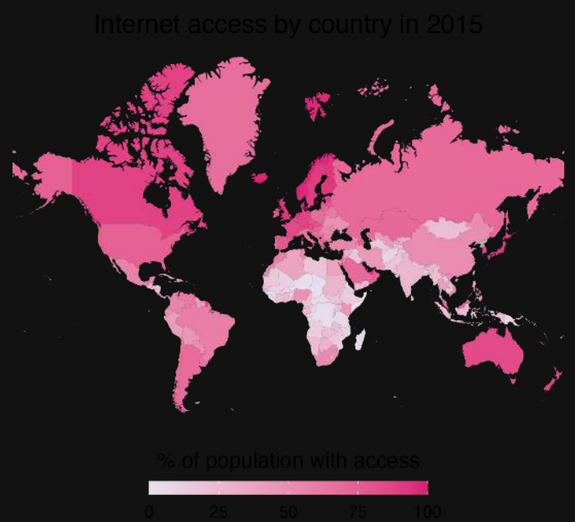
- The `iso_a3` variable in the `worlds` data frame corresponds with the `code` variable in the `internet_users_country.csv` data frame (use this for joining).
- Use `scale_fill_gradient()` to fill the color:

```
scale_fill_gradient(
    low = "#e7e1ef",
    high = "#dd1c77",
    na.value = "grey70",
    limits = c(0, 100))
```

Robinson Projection



Mercator Projection



Remember: Mini Project 3 Due 11/22

https://eda.seas.gwu.edu/2026-Fall/mini/3-redesign.html

Extra practice

Your turn

15:00

Use the `us_states_cont` data frame and the `state_abbs` data frame to create a labeled map of the U.S.:

